

NERC

NORTH AMERICAN ELECTRIC
RELIABILITY CORPORATION

DRAFT Reliability Guideline

Risk Mitigation for Emerging Large Loads

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RELIABILITY | RESILIENCE | SECURITY



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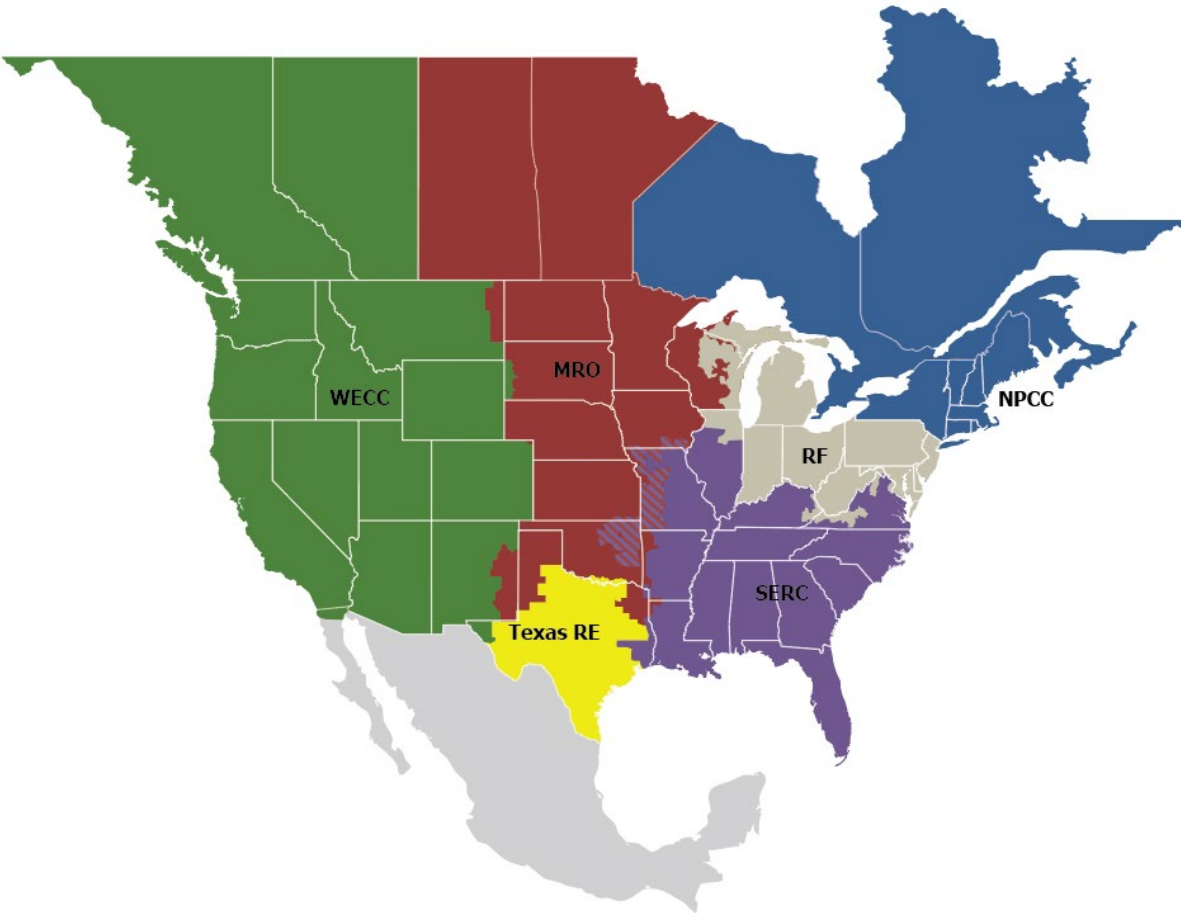
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Preface

Electricity is a key component of the fabric of modern society and the Electric Reliability Organization (ERO) Enterprise serves to strengthen that fabric. The vision for the ERO Enterprise, which is comprised of the NERC and the six Regional Entities, is a highly reliable, resilient, and secure North American bulk power system (BPS). Our mission is to assure the effective and efficient reduction of risks to the reliability and security of the grid.

Reliability | Resilience | Security
Because nearly 400 million citizens in North America are counting on us

The North American BPS is made up of six Regional Entities as shown on the map and in the corresponding table below. The multicolored area denotes overlap as some load-serving entities participate in one Regional Entity while associated Transmission Owners/Operators participate in another.



MRO	Midwest Reliability Organization
NPCC	Northeast Power Coordinating Council
RF	ReliabilityFirst
SERC	SERC Reliability Corporation
Texas RE	Texas Reliability Entity
WECC	WECC

Preamble

The NERC Reliability and Security Technical Committee (RSTC), through its subcommittees and working groups, develops and triennially reviews reliability guidelines in accordance with the procedures set forth in the RSTC Charter. Reliability guidelines include the collective experience, expertise, and judgment of the industry on matters that impact BPS operations, planning, and security. Reliability guidelines provide key practices, guidance, and information on specific issues critical to promote and maintain a highly reliable and secure BPS.

Each entity registered in the NERC compliance registry is responsible and accountable for maintaining reliability and compliance with applicable mandatory Reliability Standards. Reliability guidelines are not binding norms or parameters nor are they Reliability Standards; however, NERC encourages entities to review, validate, adjust, and/or develop a program with the practices set forth in this guideline. Entities should review this guideline in detail and in conjunction with evaluations of their internal processes and procedures; these reviews could highlight that appropriate changes are needed, and these changes should be done with consideration of system design, configuration, and business practices.

Executive Summary

Introduction

The first NERC LLTF work item, *White Paper: Characteristics and Risks of Emerging Large Loads*,¹ defines large loads—including data centers, cryptocurrency mining facilities, hydrogen electrolyzers, manufacturing facilities, and arc furnaces—as “[a]ny commercial or industrial individual load facility or aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS due to its demand, operational characteristics, or other factors.” The paper identified key risks from emerging large loads along with the characteristics that can cause those risks.

Emerging Grid Challenges

Recent developments in emerging large loads and their integration into the bulk power system (BPS) have introduced new risks to BPS reliability. Some of these risks, like customer-initiated load reduction (CILR) events and large load oscillation events, have been clearly demonstrated and observed. These events can transpire in a matter of seconds, leaving real-time operators little to no time to respond. Additionally, planners anticipate long-term risks related to transmission planning and resource adequacy. While these risks are not yet fully realized, they require proactive planning to mitigate.

Ensuring Consistent and Reliable Operations for All Users

Guidance is necessary for the reliable integration of large loads into the BPS. This reliability guideline provides actionable BPS reliability-enhancing guidance for both existing NERC registered entities as well as non-registered large load entities and original equipment manufacturers (OEM).

Reliable integration of emerging large loads is not only essential for protecting the BPS, but it also enables the system to accommodate additional large loads more quickly. When interconnection and operational processes are structured to ensure reliability from the outset, there are no reliability-based barrier to energizing new demand, allowing large loads to connect without unnecessary delay or compromise.

Guidelines as a Supplement and Bridge to Standards

Although the guidance is voluntary and non-binding, this guideline will likely heavily influence future Reliability Standards, which contribute to a more reliable and secure grid. Reliability guidelines are reviewed and updated at least every three years in accordance with the RSTC Charter and may be updated earlier in response to comments or lessons learned.

The guidance in this paper is intended to supplement, rather than replace, any existing or new Reliability Standards. It should be used in conjunction with any modifications to Reliability Standards or registration criteria changes in the NERC Rules of Procedure. This guideline is being developed concurrently with additional efforts by the LLTF to facilitate the timely and efficient integration of Large Loads into the BPS.

From Passive Consumption to Active Participation

This guideline recognizes that owners and operators of end-use facilities and equipment were not historically required to be as active in practices and processes that ensure BPS reliability. However, due to the scale and urgency of the risks to BPS reliability, it is strongly recommended that large load entities (if they are not already doing so) participate actively in practices and processes that enhance BPS reliability. The LLTF values the participation of large load entities in its task force and the resulting work products.

¹ *White Paper: Characteristics and Risks of Emerging Large Loads*. Available: https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Whitepaper%20Characteristics%20and%20Risks%20of%20Emerging%20Large%20Loads.pdf

High-Level Recommendations

Table E.1: High-Level Recommendations for Large Load Risk Mitigation

Recommendation	Applicable Entities
Applicability: The recommendations in this guideline should be applied to all large load entities interconnecting to or otherwise impacting the BPS.	Distribution Providers (DP), Transmission Owners (TO), Transmission Operators (TOP), Transmission Planners (TP), Planning Coordinators (PC), Balancing Authorities (BA), Reliability Coordinators (RC), Resource Planners (RP), large load entities, and large load equipment manufacturers
Data Collection and Modeling: Large load entities should provide, and TO/DPs should collect, site-specific modeling information for supporting system studies as part of the interconnection requirements, including the following: - Detailed load characteristics and information as required by TOs, TPs, and PCs for developing power flow and dynamic models. - Any data or modeling information (e.g., load forecasting) as specified by the RCs, BAs, or TOPs to adequately, reliably, and in a timely fashion perform their required analysis (similar to TOP-003 and IRO-010 for Generator Owners (GO)). Large load entities should provide, and TO/DPs should collect, accurate, verified, updated dynamic modeling data (as-planned, as-built, and as-left), comparable to current generator dynamic model efforts (e.g., Project 2020-06) to ensure accuracy and validation of dynamic modeling data.	DPs, TOs, TOPs, TPs, PCs, BAs, RCs, RPs, large load entities, and large load equipment manufacturers
Interconnection Requirements and Processes: TOs and DPs should incorporate TP, PC, RC, TOP, or BA-specified requirements necessary to maintain reliability in their area into their interconnection requirements (under FAC-001). TOs should establish clear design and performance criteria for end-user facilities, such as large loads, addressing identified risks, such as voltage and frequency ride-through, protection settings, stability, operational ramping limits, power quality, communication, undervoltage load shedding (UVLS), underfrequency load shedding (UFLS) and manual load shedding (MLS) participation, high-resolution data monitoring, and oscillation mitigation. TPs and PCs should require large loads to submit detailed load characteristics and modeling information and specify the type of load (e.g., paper mill, steel mill, data center) as part of the interconnection process.	DPs, TOs, TOPs, TPs, PCs, BAs, RCs, RPs, large load entities, and large load equipment manufacturers
Interconnection Study and Coordination Process Enhancements: TPs and PCs under TPL-001 should perform comprehensive transmission planning using as-planned steady-state and dynamic models to assess reliability impacts that consider the unique characteristics of large loads.	DPs, TOs, TOPs, TPs, PCs, BAs, RCs, RPs, large load entities, and large load equipment manufacturers

Table E.1: High-Level Recommendations for Large Load Risk Mitigation

Recommendation	Applicable Entities
This includes consideration of individual and collective ride-through behavior during system disturbances or faults. TPs and PCs should also assess frequency stability including overfrequency performance, the need for overfrequency protection schemes, and the Most Severe Single Contingency (MSSC) on the demand side. Large load entities, TOs, and DPs should coordinate under EOP-011 and FAC-001 interconnection requirements to ensure that they are designed and perform appropriately when included in UVLS and UFLS schemes or MLS plans of DPs, TOs, or TOPs.	
Real-Time Monitoring and Performance Evaluation: TOs are advised to install high-resolution monitoring equipment (such as advanced digital fault recorders (DFR) and Phasor Measurement Units (PMUs)) and relevant data streaming/archiving infrastructure as part of their interconnection requirements at interconnecting substations to support oscillation detection based on the specific needs of RCs, BAs, and TOPs. TOs and RCs should revise their oscillation monitoring/mitigation platforms and procedures to account for potential load-induced oscillations. BAs and RCs should specify any interconnections design requirements to TOs in their footprint for inclusion in TO's interconnection requirements (FAC-001) to maintain grid reliability, such as operational ramping limits, interconnection/integration process requirements, communication protocols, and monitoring equipment. BAs, TOPs, and RCs should specify any data requirements to large loads (or TOs/DPs if large loads are not registered) necessary to perform operational and real-time analysis and monitoring, such as models and load forecasting. The existing Reliability Standards (PRC-002 and PRC-028) are deemed sufficient but should be considered for modifications to include large loads if they become registered entities. BAs should ensure that they identify and incorporate their MSSC for loads (including consideration of clustered load loss due to voltage-sensitive ride-through settings for a single contingency) into their reserve considerations, comparable to Generation MSSC (BAL-002). Large loads (or TOs/DPs if large loads are not registered) should ensure that they respond and adhere to all Operating Instructions as instructed (TOP-001, IRO-001). Additionally, the entity responsible for adhering to Operating Instructions at customer sites must be trained appropriately (PER-005)	DPs, TOs, TOPs, TPs, PCs, BAs, RCs, RPs, large load entities, and Large Load OEMs

Table E.1: High-Level Recommendations for Large Load Risk Mitigation

Recommendation	Applicable Entities
Communication Requirements: Large load entities, in coordination with their interconnecting TO or DP, should be required to establish Interpersonal Communication capabilities (like COM-001 for Generator Operators (GOP)) for large load facilities and operations to maintain grid reliability during normal and emergency conditions.	DPs, TOs, TOPs, TPs, PCs, BAs, RCs, RPs, large load entities, and large load OEMs
Cyber Security Information Protection (CIP) and Cyber Security Requirements: Large load entities should be provide adequate, reliable protection of their assets and any Bulk Electric System Cyber System Information (BCSI).	Large load entities, and large load OEMs
Resilience, System Restoration, and Load Shedding: TOs should coordinate with PCs to determine if it is appropriate to include the large loads in their UFLS/UVLS programs.	TOs, PCs

Introduction

Purpose

This reliability guideline shares recommended practices, technical insights, and lessons learned for maintaining BPS reliability while integrating emerging large loads. This guideline will suggest mitigations for risks discussed in the prior LLTF white papers.

Applicability

This reliability guideline is intended for the following NERC registered entities, external entities, and broader groups:

- Planning Coordinators (PC)
- Transmission Planners (TP)
- Transmission Owners (TO)
- Transmission Operators (TOP)
- Distribution Providers (DP)
- Balancing Authorities (BA)
- Reliability Coordinators (RC)
- Resource Planners (RP)
- Large Load Entities (developers, owners, operators)
- Load-Serving Entities (LSE)
- Original Equipment Manufacturers (OEM) for Large Load Facility Equipment
- Generator Owners (GO)
- Generator Operators (GOP)
- Reliability and Security Technical Committee (RSTC) subgroups

Background

The first NERC LLTF work item, *White Paper: Characteristics and Risks of Emerging Large Loads*, defines large loads—including data centers, cryptocurrency mining facilities, hydrogen electrolyzers, manufacturing facilities, and arc furnaces—as “[a]ny commercial or industrial individual load facility or aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS due to its demand, operational characteristics, or other factors.” The paper identified key risks from emerging large loads along with the characteristics that can cause those risks.

The second NERC LLTF work item, *White Paper: Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads*, explains where there are gaps in addressing risks from emerging large loads in existing practices and Reliability Standards. The paper also categorizes risks based on impact and likelihood to give recommended actions for mitigation. Mitigations include Reliability Standards, NERC alerts, and reliability guidelines. This reliability guideline addresses the prior paper’s recommendation to mitigate certain risks with a reliability guideline.

This guideline moves beyond identification of risks and gaps to provide actionable, entity-specific recommendations. It should be read as both a supplement to and a bridge between the white papers and potential future Reliability Standards. Reliability guidelines are just one form of mitigation employed by NERC’s Framework to Address Known

and Emerging Reliability and Security Risks.² Reliability guidelines are “the common approach to use when addressing moderate impact sustained risks that are unlikely, and low impact sustained risks that are unlikely or likely.” Reliability guidelines may also be used “to establish performance expectations for emerging risks prior to codifying those expectations into Reliability Standards.” This reliability guideline will be used for both purposes: tackling the high-impact high-likelihood risks identified in the LLTF’s prior white paper while also addressing moderate- and low-impact risks.

Vocabulary Terms

The reader of this reliability guideline should note that all terms contained in the NERC *Glossary of Terms*³ and used in this guideline are capitalized. In addition to those terms, some additional terms are defined below. Terms in italics are distinguished from those defined and approved by NERC.

Area Control Error (ACE): The instantaneous difference between an entity’s Actual Net Interchange (NIA) and Scheduled Net Interchange (NIS), taking into account the effects of Frequency Bias, of correction for meter error, and of Inadvertent Interchange Management (IIM) if operating in the IIM mode. For compliance usage, refer to the term Reporting ACE.

As-built: The parameters and layout of a facility as it was actually constructed and installed, including any deviations from the original design.

As-designed: The planned parameters and layout of a facility, based on preliminary engineering design prior to construction or installation.

As-left: The configuration and parameters of a facility following commissioning and tuning, representing its final operational state.

Automatic Generation Control (AGC): A process designed and used to adjust a Balancing Authority Area’s Demand and resources to help maintain the Reporting ACE in that of a Balancing Authority Area within the bounds required by applicable NERC Reliability Standards.

Behind-the-Meter (BTM) Generation: Generation resources that are connected to the BES through the same point of interconnection as a load.

BES Cyber System Information (BCSI): Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BCSI may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.

Co-Location: Connection of two distinct elements behind the same point of interconnection.

² “Framework to Address Known and Emerging Reliability and Security Risks,” NERC, Feb. 2021. Available: https://www.nerc.com/comm/RISC/Related%20Files%20DL/Framework-Address%20Known-Emerging%20Reliability-Security%20Risks_ERRATTA_V1.pdf

³ “Glossary of Terms used in NERC Reliability Standards,” NERC, July 2025. Available: https://www.nerc.com/pa/Stand/Glossary%20of%20Terms/Glossary_of_Terms.pdf

Cryptocurrency: A class of digital assets that function in a medium of exchange, unit of account, and/or store of value, recorded and transferred on cryptographically secured, distributed ledgers (blockchains).

Data Center: A physical room, building, or facility that houses information technology (IT) infrastructure for building, running, and delivering applications and services.

Digital Fault Recorder (DFR): A device that samples the analogue values of voltage and current and converts them to a digital format after being triggered by events or signals from protection relays.⁴

Electrolyzer: A device that uses electricity to split water into hydrogen and oxygen.⁵

Electromagnetic Transients (EMT): A power systems domain that considers time-domain instantaneous values rather than phasor domain positive-sequence values for highly detailed analysis.

Energy Reliability Assessment (ERA): Assessment of the resources necessary to reliably supply the Electrical Energy required to serve Demand and to provide Operating Reserves for the Bulk Power System throughout the associated assessment period.

Large Load: Any commercial or industrial individual load facility or aggregation of load facilities at a single site behind one or more point(s) of interconnection that can pose reliability risks to the BPS due to its demand, operational characteristics, or other factors. Examples include, but are not limited to, data centers, cryptocurrency mining facilities, hydrogen electrolyzers, manufacturing facilities, and arc furnaces.**Error! Bookmark not defined.**

Model Validation: The process of comparing simulation results with measurements to assess how closely a model's behavior matches the measured behavior.

Model Verification: The process of confirming that model structure and parameter values represent the equipment or facility design and settings by reviewing equipment or facility design and settings documentation.

Parameter verification reports: A file or other evidence that demonstrates model parameters accurately reflect the real equipment and settings modeled.

Phasor Measurement Unit (PMU): A time-synchronized measurement device that estimates the magnitude and phase of an electrical phasor quantity like voltage or current.

Point of Interconnection: The location of electrical connection between one system and another (i.e., from a generator or load to the transmission system).

Price Responsive: Behavior of a load facility where demand is dependent, in part or in whole, on energy prices.

Ramp Rate*: The rate, expressed in megawatts per minute, that a generator or load changes its output or demand. *Ramp Rate is defined in the NERC *Glossary of Terms*, but it excludes mention of demand changes.

Registered Entity: Entities required by law to register with NERC and comply with NERC Reliability Standards.

⁴ Digital Fault Recorder, Entsoe.eu, Mar. 25, 2025. <https://www.entsoe.eu/technopedia/techsheets/digital-fault-recorders/>

⁵ U.S. Department of Energy, *Hydrogen Production: Electrolysis*, Energy.gov, 2024. <https://www.energy.gov/eere/fuelcells/hydrogen-production-electrolysis>

Site control: The legal authority or permission to occupy, develop, or use a specific parcel of land or property.⁶

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⁶ *Site control: Overview, definition, and example*, Cobrief.app, Mar. 29, 2025. <https://www.cobrief.app/resources/legal-glossary/site-control-overview-definition-and-example/>

Chapter 1: Data Collection, Sharing, and Monitoring

Data Collection and Sharing

Data collection is foundational to ensure that the BPS integrates large loads in a manner that supports continued reliability. Collecting data about load facilities helps grid planners and operators make informed decisions about planning and operating the BPS in a way that ensures reliability. Without the proper data to simulate and characterize large loads, grid planners and operators will be unable to accurately model large loads, leading to inaccurate analysis of BPS security and uninformed decisions when planning and operating the grid. This means that grid planners and operators need to specify the type and format of data required from large load entities across the interconnection, planning, and operations domain.

The second critical component to ensure reliability and mitigate risks after collecting the data to simulate large loads is to share the data with relevant entities. Once the interconnecting entity (i.e., TO/DP) collects data about the end-use facility, it has a duty to share that data with the appropriate entities (e.g., TP, PC, RC, BA, and TOPs) that rely on that data to ensure reliability. Many NERC Reliability Standards already exist for this exact purpose (e.g., IRO-010 and TOP-003).

This chapter aims to specify beneficial data collection and sharing activities and processes that will support planning and operational mitigations to effectively and efficiently mitigate risks to BPS reliability. The data sharing guidance in this reliability guideline is meant to supplement that guidance, not replace it.

Data Collection Stages

Load interconnection processes and stages often vary from entity to entity. This reliability guideline will break down data collection into three main stages, as shown below in [Figure 1.1](#): interconnection evaluation, integration and commissioning, and operations.

The interconnection evaluation phase spans from when the TO/DP first receives the request to interconnect to its Facilities until the interconnection agreement is signed. During the evaluation phase, the load facility may not yet have a complete design or have procured the physical equipment that will make up the load facility.

The integration and commissioning phase starts when the interconnection agreement is signed and ends when the load facility is first energized with utility power. During this phase, the load facility finalizes its design and constructs the load facility.

The operations phase starts when the load facility is energized with utility power and ends when the facility is fully retired. During operations, grid operators and planners expect to receive key operational data on an ongoing basis and following any site modifications, upgrades, or other qualifying changes.

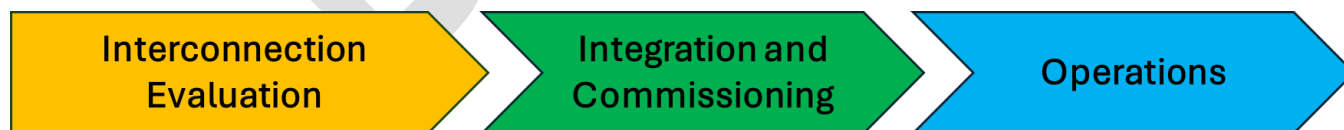


Figure 1.1: Data Collection and Modeling Throughout the Project Lifecycle

When a large load seeks to connect to the grid, utilities are required to perform a full reliability study per FAC-002. These studies are performed to identify any potential impacts to the transmission system along with any network upgrades that may be required to reliably integrate these large loads into the BPS. Due to the complexity and time required to perform these studies, many utilities may have their own process to conduct the steady-state and transient stability evaluation at various stages of the interconnection process, as shown in [Figure 1.1](#).

Large load entities, TOs, and DPs should collaborate with TPs and PCs through the TO's/DP's interconnection requirements per FAC-001 to provide all the necessary information required during various phases of the interconnection process. This collaboration will ensure the provision of site-specific data necessary for developing accurate models for steady-state and dynamic studies as part of the interconnection requirements. Due to the risk of inaccurate or incomplete data submission, some utilities may require large load entities to prepare and submit the models and supporting documentation for review. Alternatively, other utilities may develop the models themselves by using the information provided by the large load entity. Both approaches have inherent risks: customer-submitted models may lack transparency or independent validation while utility-developed models could miss critical details if customer data is incomplete. Standardized procedures for model verification, model validation, and attestation of large load data/models are recommended to close this gap.

Interconnection Evaluation Stage

During the evaluation stage, the large load typically only has access to preliminary design plans and does not complete any construction. This preliminary design data will form the basis of BPS security assessments by the interconnecting entity (TO or DP) until design plans are finalized in the next stage and the assessments can be refined with data based on actual procured equipment parameters. This stage focuses on getting as much data as possible while recognizing that the final design may be different from what was originally studied. TOs, in coordination with TPs and PCs, should create interconnection request requirements that establish a minimum set of data and milestones that must be achieved prior to signing an interconnection agreement. This minimum data requirement should allow for the large loads to be fully and properly studied per FAC-002. These studies are explained further throughout the rest of the reliability guideline. Recommended minimum data requirements for the interconnection study process are as follows:

- Steady state
 - Requested peak net real power demand
 - This includes load facility ramp up schedule (MW requested to energize on quarterly/yearly basis)
 - Overall large load power factor (value and lead/lag) information for various loading (% rated load) levels
 - Preliminary detailed one-line diagram of the substation connecting to the BES point of interconnection showing network topology, breaker configurations, etc.
 - Reactive devices (STATCOM, switched shunt, etc.)
- Dynamic
 - As-planned phasor domain dynamic load model that includes at least the following elements:
 - Protection and Control
 - Disclosure of all protection devices, OEM test reports, and equipment settings
 - Disconnection and real-power reduction thresholds for voltage and frequency events (levels and duration) at which the components within the large load will disconnect from the utility system or transfer to backup system
 - Reconnection threshold (levels and duration) at which the components within the large load (e.g., IT load, cooling load) will reconnect to the utility system and real-power ramp rates
 - Device dynamics
 - Constant voltage, current, or power behavior
 - Active power electronics
 - Reactive devices (STATCOM, switched shunt, etc.)
 - Any hardware or software solutions in the load end to smooth out oscillations

- Demand Behavior
 - Ramp rate expectations
 - Maximum ramp rates (up and down)
 - To the best extent feasible, sub-hourly profiles should be collected
 - Load make up/usage information, such as portion of AI training load in data center loads
 - Any cyclical demand ramps and their associated frequencies (especially in the sub-synchronous frequency range)
- EMT domain model (as required by TP and/or PC)
- Short circuit
 - For all applicable steady-state elements:
 - Positive, negative, and zero sequence data.
- Harmonics
 - Expected current and voltage harmonic injection spectrum from the facility and aggregated at point of connection
- Other
 - Backup generators not disclosed through the regular generator interconnection process
 - Attestation whether the large load's demand is responsive to signals, such as energy prices or emissions. Examples include:
 - Cryptocurrency mining large load facility that changes demand based on strike price of Bitcoin and real-time price of electricity
 - Data center that reduces net demand using BTM generation during real-time energy price spikes
 - Energy prices, emissions, commodity prices, cryptocurrency prices, etc.

Integration and Commissioning Stage

During the integration stage, the large load entity finalizes the load facility design and performs construction. During this phase, the large load entity should be providing finalized, verified, as-built data to the TO. In turn, the TO should share the data updates with TPs, PCs, and other relevant entities. Sharing the latest updated data is especially important for data like dynamic models. In this stage, the large load entity should provide as-built data to the interconnecting entity (TO or DP).

- Steady state and short circuit
 - Update previously mentioned data to reflect as-built facility
- Dynamics
 - Updated as-built phasor domain dynamic load model that includes at least the following elements:
 - Updates for all prior mentioned attributes
 - Model verification and attestation of model parameters
 - Rated total active power (MW) consumption of different large load components (e.g., IT load, lighting load, power distribution, and cooling load)

- A detailed breakdown of load makeup, including IT equipment, motor load, VFD, power electronics, and lighting, including attestation of model parameters that map to load makeup
- Other
 - Updated information about signals that will affect demand at the large load facility
 - Communications
 - Real-time large load entity contact for emergencies

Operations Stage

The operations stage is defined by ongoing coordination and data sharing between the large load entity, TO, TOP, TP, PC, BA, and RC. This stage is broken into two main categories: ongoing data sharing (telemetry of equipment statuses and measured values, disturbance monitoring and reporting, etc.) and data updates (new dynamic model, additional load, etc.). Per FAC-002, qualified changes to the large load facility must be studied. In addition, FAC-002 states that PCs must maintain a publicly available definition of qualified change. It is strongly recommended that PCs include any changes to demand over a determined threshold or dynamic model performance in their definition of qualified change. Updating data that was provided in prior stages is crucial to ensuring ongoing reliable operations of the BPS. However, the operations stage also involves ongoing event- and time-based data collection and coordination activities as well as ongoing real-time monitoring. In concert, all of these ongoing data collection activities will support required studies that ensure reliable operation and planning of the BPS. Large load entities should provide and TOs should collect and share with relevant entities at least the following:

- Data Updates
 - Any updates for as-built and as-left facility information, including all prior mentioned steady state, dynamics, and short circuit data.
 - Coordinate with TP and PC to provide the updated modeling information that reflects any changes made due to addition of incremental load, parameter, and/or settings changes during commissioning for planning studies (as-left).
 - Provide all necessary information needed for the BA, TOPs, RCs to conduct real-time analyses and assessments (TOP-003 and IRO-010) to address the potential risks due to large load ramping, disconnection, and reconnection to the grid. Considerations should be given to expected load profiles, equipment specifications, ramp rates, load forecasts, real-time telemetry, event data, and logs following grid disturbances, backup generation, demand response, etc.
 - Submit the most current data to TOs and/or TOPs, through the TO's interconnection requirements in FAC-001 or TOP's operating agreement. Large loads should submit all requested data according to the formats and update frequencies specified in the interconnection and operating agreements.
 - Provide updated models whenever qualified changes are made to the facility.
- Ongoing Data Sharing
 - Real-Time Data Sharing at Point of Interconnection
 - Telemetry of POI voltage, real power, and reactive power
 - Telemetry of status (ramp up, normal operations, ramp down, de-rate, etc.)
 - Near-Term Data Sharing
 - Day-ahead operating plan
 - Outage scheduling information
 - Monitoring Equipment (Real-Time and Off-line) at Point of Interconnection

- Supervisory control and data acquisition (SCADA) systems to provide telemetry on the facilities' real and reactive power consumption to the TO, BA, RC, and TOP
- PMU (better suited for oscillations < 10 Hz) with real-time data streaming to TOP, BA, and RC for real-time monitoring and post-event analysis
- DFR with offline data storage (at least three years of retention) for dynamic model validation and performance evaluation
- Continuous point-on-wave measurement device with offline data storage (retention period to be determined by TP and PC) for oscillation analysis and potentially additional dynamic model validation
- Post-Event Data Sharing
 - Share all required event data with relevant entities upon request within 30 days of request

Model Verification and Validation

Accurate dynamic models of large load facilities are required to understand reliability implications and potential mitigations. The inaccurate modeling of large loads can be mitigated by verifying the models and modeling information at various stages of the interconnection process, followed by validation processes during and after commissioning processes.

Large load entities, in coordination with the OEMs of the facility equipment, should create and maintain validated as-left models so that grid planners and operators can make informed decisions regarding BPS operations. Therefore, large load entities should supply (and TOs/DPs should require) accurate, verified, and updated data at various stages (e.g., as-designed, as-built, and as-left) of the planning, integration, and ongoing operations phases of the project lifecycle, including any material updates to equipment that would impact modeling or settings. For the verification and validation process the following strategies are recommended:

- **Model Verification Process**
 - Large load entities should submit and maintain dynamic models that accurately represent the installed equipment and parameters. This includes performing model verification and providing the model verification report to relevant entities (TPs and PCs) at some point prior to energizing.
 - Large load entities should provide attestation that the as-built model parameters represent the installed equipment.
 - Each TO, in coordination with its TP and PC, should establish large load model quality requirements.
- **Model Validation and Performance Analysis**
 - **Develop Event Data Collection Processes:** Create a new NERC standard or modify existing ones (e.g., PRC-002, PRC-028) to install high-resolution metering and digital fault recording for large loads, specifying data resolution and format.
 - **Establish Model Validation and Post-Event Analysis Process:** TPs and PCs should develop a process for post-event data sharing and analysis, involving large load entities, TPs, TOPs, BAs, and RCs, to validate the large load's performance and model accuracy based on the collected measurement data.
 - **Establish Process for Model Updates and Restudy:** TPs and PCs should establish a process for large load models to be updated given model validation and post-event analysis data. They should also establish criteria for model updates and what changes require a restudy.

Chapter 2: Interconnection

Interconnection Process

TPs and PCs should establish a comprehensive interconnection process and periodic near-term system-wide study process using steady-state, dynamic, and short-circuit models to assess the reliability impacts of large loads. TPs and PCs should coordinate with TOs and DPs in their areas to collect data needed for the interconnection process.

TOs, in coordination with TPs and PCs, should establish a comprehensive interconnection process that includes the following:

- A method for collecting, at a minimum, the data required in [Chapter 1](#)
- A definition of readiness criteria and an evaluation of any new interconnection application against the readiness criteria before continuing in the interconnection process
- Interconnection study agreement and associated process
- Interconnection studies using steady-state, dynamic, and short-circuit models to assess reliability impacts of large loads
- A periodic process in the near term that studies the collective impact of all large loads in a region defined by the PC that energize in a common timeframe using steady-state and dynamic models
- A comprehensive commissioning process that ensures operational readiness for large loads, as detailed in [Chapter 4](#)

Interconnection Studies

TPs and PCs should incorporate at least the following in their large load interconnection study processes:

- Steady-state studies based on the as-planned data provided by large loads to the TOs and updated as necessary, based on design, upgrades, or other qualifying changes
- Transient dynamic analyses that consider the following:
 - The large load facility on-fault and post-Disturbance frequency and voltage recovery behavior
 - The collective impact of electrically close facilities, including large loads, with similar performance modes
 - The total potential magnitude of load loss considering the individual and collective tripping or customer-initiated load reduction of large loads during System Disturbances
- Transmission protection and control coordination studies that include:
 - Recommendations to TOs regarding any transmission protection coordination changes to transmission relays or modifications to large load fault interruption devices to ensure reliable isolation of system faults
 - Recommendation on changes to protection and control settings of local protective devices
 - Recommendations on changes to operation, protection, and control settings of the large loads to minimize customer-initiated load reduction during System Disturbances
 - Inclusion of large load entity input into the TO coordination of protection and control devices to identify how the large load can assist in the reliable isolation of system faults
- A process to determine the necessity to carry out special assessments, such as electromagnetic transient (EMT) studies for a given interconnection request. For example, this process could be carried out by the TP and PC when large loads interconnect to areas with weak system strength, when the large loads interconnect to large grid connected generation sources, or when the large loads interconnect to areas electrically close to other sensitive dynamic active resources.

- TPs and PCs should conduct small-signal, sub-synchronous, and transient stability studies
- TPs and PCs should develop EMT screening criteria where needed (e.g., for sub-synchronous interaction). Screening criteria specific to large loads are not yet widely established. Traditional EMT screening criteria for oscillation-related reliability issues associated with power electronics include close electrical proximity to the following:
 - Large power electronic devices, such as HVdc lines, large electric arc furnaces, inverter-based resources (IBR), or static reactive power devices.
 - Weak grid conditions. In the context of converter-driven stability, this could be considered to correspond to an short-circuit ratio (SCR) of 2 or less (as defined by the three-phase bus fault current available from the system and the total apparent power rating of the power electronics on site).
 - Series capacitors.
 - Large steam, gas, or hydroelectric turbine generators. These are of particular concern if the load profile has significant fluctuations in the 5 to 60 Hz range.

Maturity Assessment for Large Load Inclusion in Interconnection, Planning, and Operations Studies

Throughout the interconnection process, a large load will transition from a concept or design to a physical facility in the field. The question of when the large load should move to a new phase of the interconnection process and its associated studies depends on project maturity. To aid in deciding when large load projects should be selected for inclusion into various studies, TPs and PCs should establish uniform definitions for the different phases of a large load interconnection project's lifecycle and have clear criteria that determines the maturity of a project. The different stages of maturity will provide thresholds for the large load's inclusion into the next phase of studies. This will standardize the inclusion of large loads in interconnection, planning, and operational assessments. Examples of project attributes that inform the large load facility's maturity include the following:

- Data Sharing Agreements
- Interconnection Agreements
- Financial Security Posted
- Proof of Site Control
- Construction Status
- Development Permit Status
- Environmental Permit
- Land Status
- Procured Equipment

Additionally, a maturity assessment framework will enable grid planners and operators to ensure realistic large load project modeling in their planning assumptions, thus helping to mitigate the risks posed by inaccurate long-term forecasting. Delayed inclusion of large load interconnections into a long-term planning model or set of assumptions impacts BPS reliability through invalid and ineffective transmission planning assessments and studies. Such issues often arise from speculative site selection or from insufficient information provided by large load entities. In many organizations, planning activities, such as generator interconnection, transmission upgrades, and resource adequacy are handled by different departments and usually have different planning cycles. These processes should be synchronized between various departments using the maturity assessment framework to minimize redundancy and identify issues efficiently for large load integration.

Triggers for Re-Study

Similar to the existing Reliability Standard FAC-002, TPs and PCs should create and publish criteria that triggers a re-study of the large load facility. Experts in the ERCOT region state that some large load facilities have changed from a data center to a cryptocurrency mining facility after receiving approval of their interconnection request application. Late-stage modifications like this would almost certainly require a new dynamic study. This scenario is clear, but less-clear scenarios should be examined in the PC's re-study criteria.

TOs and DPs should have clear policies for handling significant changes in load characteristics that occur after the initial interconnection studies. Large load facility characteristics may vary significantly with design decisions made during the interconnection process (e.g., the selection of specific vendors for electronic devices or determination of the site's mix of computational processes). These changes may introduce stability-related risks that were not identified in the initial studies as the result of the mismatch between prior and current load model data. This could necessitate technical mitigations (e.g., system upgrades or devices, such as energy storage or reactive power support) that require significant cost and lead time to install. If these mitigations cannot be put in place by the original energization date, it may be necessary to delay energization or restrict the load's operational range until the mitigations are in place. For almost any facility, some differences between as-planned, as-designed, and as-built models are inevitable, and these policies should provide reasonable accommodation for changes in load characteristics that do not significantly alter the reliability risks associated with the load interconnection. TOs, DPs, and load owners all face cost, schedule, and reliability risks when major changes to interconnection scope occur. Therefore, they should collaborate to identify and prioritize specific model data and design decisions that are most crucial for accurate BPS security assessments as early as possible.

Chapter 3: Long-Term Planning and Resource Adequacy

Long-Term Planning and Coordinated Studies

The incorporation of new large load interconnection requests into reliability assessments is not only vital to accurate studies of the BPS but also needed to ensure that future transmission design is adequately modeled. Inadequate modeling of large load interconnections poses risks to BPS planning by potentially leading to incorrect study results and incorrect long-term planning assumptions.

The primary risk to BPS reliability from large loads in the long-term planning and resource adequacy domains is insufficient transmission and generation caused by under-forecasting demand. If under-forecasting leads to probable transmission or resource adequacy issues, then the TOs, TPs, PCs, BAs, and RCs should have a process for addressing those risks. One example of mitigation for that risk is establishing a system operating limit (SOL) or Interconnection reliability operating limit (IROL).

Regardless of whether large load studies are performed in a cluster or serially, TPs and PCs should coordinate with TOs to establish study processes to ensure that large load facilities interconnected to the BPS are adequately studied.

Analysis for Transmission Planning

The sudden and uncoordinated loss of a large load could cause overvoltage and overfrequency conditions that trigger protective relays that lead to a cascading outage, and the highly variable nature of some large loads profile can induce risks outside of the traditional peak and off-peak scenarios studied in transmission planning analysis. Therefore, when conducting analysis for transmission planning studies, TPs should modify their study assumptions to capture the risks that large loads pose due to their size and operational behaviors.

To mitigate the risks posed by the sudden loss of a large load, the loss of the large load should be studied as a contingency in both steady-state and dynamic assessments by the TP and PC. Similarly, the TP and PC should study the loss of large loads clusters (all in close electrical proximity to a single fault or disturbance) and provide that information to the RP, BA, and TOP.

Planning processes should analyze various loading scenarios representing a diversity of load demand and generation patterns as appropriate for their areas in both steady-state and dynamic assessments as well. The dynamic assessment should include voltage stability and frequency stability assessments as documented in [Chapter 5](#).

Interregional Coordination Between Grid Planners

If a TP or PC studies and approves a project, such as an addition of a transmission line, that project can significantly impact the neighboring entities depending upon the size and location of the project. Therefore, TPs and PCs should coordinate with neighboring entities to account for such projects, whereby the models, the contingencies, and the study results are shared with the neighboring entities to help them perform their own studies.

Large loads interconnecting at the seam/boundary with a neighboring entity could have significant reliability impacts on that entity. Therefore, it is important for the RCs, TPs, and PCs to establish the processes for sharing pertinent information and data of the large load facility, such as load size, type, dynamic model data, and protection settings, with the neighboring entity at the interconnection stage. This allows the neighboring entity to perform its own studies whether independently or in collaboration with the incumbent entity. Moreover, there should be processes for the entities to share the results of reliability studies where any reliability issues are found at the neighboring entities' facilities. This would facilitate identifying system upgrades to mitigate such reliability issues.

Resource Adequacy Analysis

Given that large loads frequently seek connection to the grid faster than transmission can be built and due to the size of these large loads, they can pose risks to resource adequacy as documented in the NERC LLTF *White Paper: Characteristics and Risks of Emerging Large Loads* section on Resource Adequacy.

Simulation of Delayed Generation Interconnections

Demand may outstrip generation supply as large loads are added faster than the addition of generation resources. These large loads are expected to require firm service, limiting their ability to be considered flexible or providing limited demand-response capabilities. This directly impacts BPS reliability through the reduction of BA Operating Reserves. To address this, RPs should model scenarios in their evaluation of resource adequacy that incorporate accelerated large load additions coupled with delayed generation additions to assess worst-case scenarios.

RPs should coordinate with load-serving entities, TOs, DP, and TP/PCs to identify the expected large load growth to incorporate in their studies, consistent with state, provincial, and federal regulatory requirements. The modeling criteria may consider “tiers” of load additions, like the generator tier levels of the *Long-Term Reliability Assessment* (LTRA), to demonstrate the quantity and timing of large load additions based upon confidence levels for those values. RPs can use this information to understand the level of risk associated with the timing of large load additions coupled with potential delays in generation additions.

Without the ability to model the timing and likelihood of new large load additions and the timing and likelihood of new generation resources, RPs could miss key inputs to resource adequacy studies. These missed inputs could lead to inaccurate resource plans or the underestimation of the risks associated with those plans. Additionally, TOs should establish a FAC-001 process to verify with RPs that there is sufficient generation adequacy prior to interconnecting new large loads.

Simulate Operational Constraints and Flexibility of Large Loads

The risk to resource adequacy from the lack of representation of large loads’ operational flexibility and constraints comes from NERC’s LLTF *White Paper: Characteristics and Risks of Emerging Large Loads***Error! Bookmark not defined.** and NERC’s LLTF *White Paper: Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Loads*. NERC’s *Technical Reference Document: Considerations for Performing an Energy Reliability Assessment*⁷ (NERC Energy Reliability Assessment Working Group) echoes this by emphasizing the need for accurate modeling of demand response and non-conventional load behavior.

Large loads may have operational constraints, partial controllability, or internal energy assets. They (if co-located) may appear firm but exhibit flexible or unpredictable usage. They can have rapid ramp rates (up or down), operational constraints, and BTM generation or storage that traditional models do not capture. If operational characteristics are not represented in resource adequacy models and flexibility assessment, planners may overestimate available flexibility or underestimate load variability—leading to misaligned resource procurement and reliability events during peak demand, large load switching on and off, or transferring to co-located generation conditions.

PCs and RCs should update their planning processes and resource adequacy models to include realistic assumptions about operational constraints and flexibility of large loads. They should follow NERC’s ERA guidance to model demand-side variability, including time-coupled constraints and state-dependent behaviors. The updated models should be able to capture dispatch constraints (e.g., minimum run times, startup/shutdown behavior), separate firm vs. flexible load components, time-varying operating limits (e.g., AI training window), and characteristics of the BTM resources (e.g., solar power plants, battery energy storage system, backup gensets), and run sensitivity analysis under scenarios (e.g., no load flexibility, partial or full demand response availability, fast ramping events).

⁷ *Technical Reference Document: Considerations for Performing an Energy Reliability Assessment*, NERC, Dec. 2024. Available: https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Technical%20Reference%20Document%20Considerations%20for%20Performing%20an%20ERA%20V2.pdf

Large load entities should provide documentation of expected dispatchability, ramp rates, internal generation/storage characteristics, and technical constraints and operating limits (e.g., IT workload cycles, min/max uptime). This mitigation ensures that load behavior is realistically modeled, preventing false assumptions about demand-side flexibility. Additionally, this mitigation improves accuracy of load/resource balance forecasts, enhances planners' ability to predict and prepare for realistic worst-case conditions, and better aligns planning decisions with actual load behavior. This mitigation enables planners to do the following:

- Accurately size reserve margins
- Procure fast-ramping resources or demand-response capacity only when needed,
- Avoid unexpected load persistence that could lead to loss-of-load events.

Mitigation is sufficient when:

- Flexibility is modeled using real operational bounds,
- The models explicitly represent load controllability attributes,
- Distinguish between firm and interruptible demand,
- Assumptions are vetted with large load entities and sensitivity tested (use data from actual load tests or validated technical documentation),
- Have sensitivity scenarios that incorporate different flexibility levels (e.g., if load is fully interruptible vs. firm).

It is important to note that most resource planning models evaluate adequacy across an integrated hourly (or minute) time frame. Some large loads may exhibit high intra-minute volatility across normal operations that would need to be considered for appropriate regulating needs. RPs should consider methods to obtain this volatility information from customers and assess risk and system needs to appropriately manage this intra-minute volatility.

The importance of this action is critical. Inadequate modeling of large loads' behavior and flexibility can result in resource shortfalls during the tightest periods, compromising BPS reliability.

Simulate Transmission Constraints in Loss of Load Expectation Analysis

Both NERC's LLTF *White Paper: Characteristics and Risks of Emerging Large Loads* and NERC's LLTF *White Paper: Assessment of Gaps in Existing Practices, Requirements, and Reliability Standards for Emerging Large Load* discuss the need to include transmission system constraints and locational risk in resource adequacy assessments. The first white paper discusses the locational nature of reliability risks, and the second white paper identifies gaps in modeling transmission constraints in resource adequacy studies. The NERC *Technical Reference Document: Considerations for Performing an Energy Reliability Assessment* emphasizes the importance of modeling transmission-limited zones, deliverability, and non-firm transfers.

Large loads can be highly concentrated in specific locations and can shift locational demand profiles. They tend to cluster geographically near favorable conditions (e.g., low-cost power, substations), which may not have sufficient local generation or transmission. If loss of load expectation (LOLE)/planning reserve margin (PRM) studies do not incorporate local transmission constraints, they may miss localized capacity risks, especially during high load + outage scenarios, resulting in undetected reliability gaps. Ignoring transmission limits in LOLE modeling can result in the following:

- Underestimating zonal load-serving risk,
- Hiding bottlenecks that delay energy delivery,
- Generating overconfidence in system-wide adequacy.

TPs, PCs, and RCs should coordinate with RPs to integrate transmission constraints associated with generating resource delivery, if any, into LOLE assessments. PCs and RCs should use network-aware ERA models as described in the NERC ERA *Technical Reference Document: Considerations for Performing an Energy Reliability Assessment*,⁷ including tools that support zonal reliability assessment, stochastic unavailability, and delivered load margin metrics.

TPs should provide current and planned constraints (e.g., congestion zones), transfer capabilities under normal and contingency scenarios (N-1, N-1-1), and weather-resilient infrastructure gap transfer capability curves for the relevant planning horizon. TPs and PCs should consider any transmission constraints associated with long-term transmission and generation outage work in both peak and off-peak seasons.

Large load entities should provide accurate siting and expansion timelines to inform demand forecasts and expected reliability requirements. They should identify location, ramp-up profiles, and reliability expectations (e.g., firm vs. interruptible interconnection) and share anticipated transmission service type (e.g., firm vs. non-firm).

This mitigation is complete under the following conditions:

- Resource adequacy simulations include at least zonal constraints aligned with actual transmission limits,
- Transmission outages and congestion risk are incorporated into LOLE simulation sets
- Transmission-constrained LOLE, loss of load hours (LOLH), and expected unserved energy (EUE) metrics are reported for each relevant planning zone,
- The models reflect correlated outages (e.g., N-1-1 or weather-related) that can co-occur with large load peaks,
- Output identifies locational loss of load risk even in regions with high overall reserve margins,
- Planners coordinate outputs with local reliability and transmission planning studies.

The importance of this mitigation is critical. Without location-aware analysis, risks may remain hidden in high-growth load pockets even when overall system adequacy appears sufficient. Without this, systems may appear resource adequate but fail under real-world locational stress.

Simulate Multiple Metrics for Resource Adequacy Assessments

The white papers identify that traditional LOLE metrics alone are insufficient for capturing the full impact of large loads and recommend adoption of expanded probabilistic metrics, such as EUE, LOLH, and conditional value at risk (CvaR). The white papers note that relying only on LOLE obscures tail risks and event severity, especially for rare but impactful combinations of large load and outages.

Emerging large loads increase variability and systemic uncertainty, which can result in extreme but low-frequency events. LOLE alone may indicate compliance while ignoring deep reliability gaps in worst-case scenarios. Furthermore, emerging large loads introduce high variability and correlation risk (e.g., multiple loads energizing during tight reserve margins or weather extremes). LOLE alone might not capture the following:

- How long load is lost (LOLH)
- How much energy is unserved (EUE)
- How bad the worst cases get (CvaR)

This leads to underestimation of system risk and underinvestment in flexible resources or infrastructure.

PCs and RCs should enhance resource adequacy studies to implement the following:

- Multi-metric assessments as recommended in NERC ERA, including the following:
 - LOLH: number of hours/year when load cannot be met
 - EUE: total MWh of unserved load
 - CVaR: average EUE in worst X% of simulations (tail risk magnitude)
- Use probabilistic tools that support thousands of weather/load/generation scenarios and provide distributions, not averages
- Evaluate non-linear risk accumulation when large loads coincide with outages or extreme weather

PCs and RCs should use results to calibrate Planning Reserve Margins and justify procurement of additional dispatchable or flexible resources.

Load forecasting entities should supply appropriate input distributions (e.g., high-resolution load forecasts, load growth uncertainty, outage probability curves).

Large load entities should provide granular load shape profiles and expected behavior under normal and extreme events, flexibility characteristics and emergency response protocols, and modeled stress-event behavior (e.g., when tripped or derated). Large load entities should also provide accurate ride-through and curtailment capabilities.

Modeling tool developers should ensure that platforms can support stochastic simulation.

This approach gives planners a clearer picture of the full distribution of reliability outcomes, especially high-consequence but low-probability events. This supports the following:

- Smarter investment in reserves and demand-side programs
- Better emergency preparedness
- More transparent planning decisions for policymakers and stakeholders

The mitigation is sufficient when the following items are true:

- Studies produce probabilistic adequacy metrics (LOLH, EUE, and CvaR) across multiple scenarios,
- Tail-event risk (e.g., CVaR of top 5% events) is quantified and evaluated,
- Results are used in planning decisions (e.g., procurement, siting, interconnection approval)
- Risk thresholds are set and used to drive decisions
- Metrics are validated against historical events or stress-tested against extreme cases

The importance of this action is critical. This enables an understanding of reliability risk under uncertainty introduced by emerging large loads. Without expanded metrics, grid planners risk being blindsided by rare, severe outages that aggregate metrics like LOLE cannot detect.

Chapter 4: Operations and Balancing

Commissioning Process

In power systems engineering, commissioning is typically the process used to bridge from a planned facility interconnection to real-time operations. Commissioning can include data verification activities, additional studies, and physical preparations of the site to enable successful and reliable BPS operations.

TOs should establish a comprehensive commissioning process that ensures operational readiness for large loads. Those requirements should include but are not limited to the following:

- Verification of communications including, but not limited to the following:
 - Testing that SCADA, PMU, and other live telemetry are sending and receiving as intended.
 - If participating in demand response, testing of demand-response signals and successful deployment of demand response.
- Prior to energization of the large load facility, ensure that the loads are properly accounted for in load forecasting and other operational modeling, like outage coordination and day-ahead studies.
- TOs, RCs, and BAs, in coordination with large load entities, should conduct commissioning tests of the large load prior to interconnection and test the load facility's ramp ability and max consumption.
- Require the large load facility to be studied in a near-term operational planning assessment that includes all large load facilities nearing their energization date. This should be coordinated with the TP, PC, and RC.
- Require comprehensive dynamic model verification prior to energization.
- Require testing of demand-response capability prior to procuring demand response services from the large load facility.

Real-Time Monitoring

To support real-time operational functions like short-term forecasting, balancing, and event analysis, entities mentioned in [Chapter 1](#) should collect real-time and post-event data as needed to support analysis and other processes that ensure reliability.

Additionally, TOPs, BAs, and RCs should require real-time data feeds from large load facilities to input to EMS applications, enhancing situational awareness.

Short-Term Forecasting

Large loads pose new challenges for short-term operational forecasting by RCs, BAs, and TOs. These loads can be highly variable and exhibit both second-to-second and minute-to-minute power oscillations along with fast ramping capabilities. If large loads are not considered in short-term forecasting, the BA could under-procure generation in real time, potentially leading to reliability issues and load shedding.

Load forecasting information from large loads is critically important for BAs to assess risks to energy adequacy under BAL-007 and to ensure appropriate unit commitment and dispatch. Therefore, TOs in consultation with BAs should do the following:

- Establish required load forecasting information needs and timing of updates
 - This may include the large load entity providing near-term hourly net-load forecast and planned outages

Load Coordination in Demand-Response Programs

The TO, RC, and BA should register large loads in demand-response programs if the large load is willing to participate and can be flexible. This will allow for the load to be managed as needed to address operational conditions. The TO, RC, and BA should require large loads to submit their current operating plans that provide a forecast for the next seven days to align with BAL-007 Near-Term Energy Reliability Assessments. Without participation in demand-response programs or the submission of current operating plans, the system operator will find it challenging to properly account for large loads in their forecasts and thus energy and ancillary service procurement and operational planning.

Operational Analysis of Load Behavior

The best source of a short-term forecast would be directly from the entity that controls the demand of the large load facility. The large load entity should provide, and the BA and RC should require the large load entity to provide, near-term demand forecasts and operating plans as well as real-time telemetry.

Once the large load has become commercial, the TO, RC, and BA should review the large load's normal operational behavior to assess how it matches against their forecasted load consumption predictions.

TOs, RCs, and BAs should monitor large loads and observe their load usage patterns to inform short-term forecasting. This will allow them to understand current and past behaviors for forecasting models.

Balancing and Reserves

To ensure the security of the BPS, the BA must ensure that load and generation are constantly balanced. Given the ability of large loads to ramp hundreds of MWs in seconds as documented in *White Paper: Characteristics and Risks of Emerging Large Loads*, this poses a challenge for BAs to maintain the load and generation balance given the highly variable load profiles. Failure to maintain this balance could lead to the collapse of the BPS.

Balancing Reserve Management

Currently, large loads often operate with limited grid visibility, no coordination with BAs, and no ride-through obligations. Their abrupt load changes or disconnections lead to system imbalance, unanticipated ACE swings, regulation reserve exhaustion, and frequency instability, especially during disturbances. Large, fast-varying loads can rapidly swing demand by tens or hundreds of MW within seconds or minutes. These swings strain balancing reserves and frequency control, especially if BAs lack telemetry or coordination with the load operators.

Rather than simply penalizing poor ACE after the fact, the BA should be obligated to maintain sufficient regulating capability and coordination to handle these demand swings while staying within CPS1/BAAL limits. This reduces the chance that an unpredictable 500 MW load surge or drop will result in ACE excursions that jeopardize frequency stability or violate reliability criteria. In essence, the BA should “bring the large loads into the loop” of frequency control (either by securing more responsive resources or by working with the load operator to smooth out changes), thereby maintaining balance and compliant performance. Requiring granular telemetry and data for large loads closes the visibility gap that currently exists.

Large load entities should do the following:

- **Provide real-time measurement data and near-term predictions**
 - real power, reactive power, and voltage at the point of interconnection
 - the load's ramp rate limits or active power management settings
 - advance notice of substantial load changes (e.g., planned step changes or daily demand profiles)
- **Share ramp rate limits**
 - Submit max ramp-up/ramp-down rates and allowable frequency/voltage operating ranges

- **Register points of contact**

- Assign technical contacts at the large load facility for 24/7 grid coordination with BA and TOP

BAs should do the following:

- Identify any “large loads” in their area (e.g., above a MW threshold and/or high ramp rate) and incorporate their expected behavior into operations. This could include the following:
 - Adjusting AGC parameters or supplemental regulation reserves to cover the load’s ramp profile, and
 - Establishing coordination protocols with the load or its serving entity to manage sudden demand changes.
- Redefine a “reportable balancing contingency event” to also cover large demand fluctuations above a certain magnitude and rate.
- Carry sufficient Operating Reserves (or participate in reserve sharing) to handle either the loss of its largest generation resource or a comparably severe increase in net load and the largest expected decrease in demand due to large load ramping or customer-initiated load reduction.
- Incorporate load volatility in reserve planning: Adjust regulating and contingency reserve needs using scenario analysis that reflects large load variability.
- Integrate telemetry into ACE calculations: Include large load ramp telemetry into BA control center EMS.
- Require load curtailment participation: Establish programs for voluntary or automated demand modulation from large loads during regulation scarcity.

TOPs should do the following:

- Enable situational awareness: Use EMS displays or tools to show large load status and ramp rates.
- Coordinate events with BAs: Notify BAs of any unplanned load deviations or outages.

In absence of these mitigations, system operators will lack real-time insight into the most volatile loads on the system, directly compromising frequency control, reserve deployment, and reliability metrics like ACE.

Large Load Ramping and Effect on Area Control Error

Large loads with highly variable demand profiles can present challenges in controlling the ACE, which can be difficult to forecast even hours or minutes in advance. Additionally, highly voltage-sensitive large load facilities can result in failure to ride-through normal disturbances, ultimately leading to rapid customer-initiated load reduction. If left unmitigated, these deviations can result in system-wide reliability issues, such as frequency excursions and eventually cascading outages.

Since large loads are not registered entities under NERC, there are limited Reliability Standards to proactively mitigate this impact on the BPS. Therefore, in the absence of reliability standards, it is important for TOs and DPAs to coordinate with BAs to address this risk. This coordination ensures that ACE deviations remain within acceptable limits that ensure compliance with BAL-002 as well as ensuring that the interconnection reserve margins are maintained.

The first step to best situate BAs to successfully manage ACE in real time is by having the necessary information from large loads to support BA analysis functions, real-time monitoring, and near-term energy reliability assessments (e.g., TOP-003 and BAL-007).

Mitigating the Impact on ACE from Load Variability

Large loads can introduce significant variability and fluctuations on a second-to-second, minute-to-minute, and hourly basis. The frequency and magnitude of these fluctuations can put significant strain on a BA to successfully manage ACE.

The ability of a BA to manage potential fluctuations of large loads is dependent on corrective actions at its disposal, considering factors, such as the following:

- AGC dispatch time intervals
- Units available for frequency regulation/AGC
- Operating Reserves available

Furthermore, most current BAL standard metrics focus on performance across 30-minute intervals or greater, which may not account for performance risks associated with highly variable ACE deviations on an intra-minute basis. If these risks are not mitigated, it puts the burden not only on the interconnected BA but on the entire interconnection.

The traditional dispatch hierarchy of a) unit commitment ensuring that adequate supply is available to meet projected needs, b) dispatch being utilized to follow load and intertie transactions at defined intervals, and c) frequency regulation leveraged to manage intra-dispatch variability may be inadequate to manage projected variability from large loads. Steps to manage impacts of variability include the following:

- Work with large load entities to minimize variability via software and hardware mitigation⁸ and devices like E-STATCOMs or battery energy storage systems that can help smooth the consumption as seen from the point of interconnection.
- TOs should consult with and incorporate any BA-specified requirements necessary to maintain ACE and reliability into their interconnection requirements. At a minimum, this should include the establishment of operational load ramp rate limits (e.g., MW/min) for large loads in normal and post-disturbance scenarios.

Mitigating the Impact on ACE from Customer-Initiated Load Reduction

High amounts of load reduction or loss for a single contingency or event can present significant challenges to BAs in maintaining ACE if scheduled regulating-down frequency regulation service is insufficient. Compounding this challenge is that large loads may self-isolate from the grid for minutes or hours before reconnecting. If the BA is forced to decommit generation to maintain ACE for the initial load loss, there may be insufficient dispatchable generation on-line or regulating-up frequency regulation service to maintain ACE when the large loads reconnect or ramp up to return to their normal or expected consumption levels.

Therefore, BAs in coordination with their TOs, TPs, and PCs, similar to Generation MSSC requirements in BAL-002, should identify and incorporate their MSSC for loads into their unit commitment and reserve considerations. The MSSC should consider clustered load loss due to the collective ride-through behavior of loads in electrical proximity for a single fault or disturbance.

TOs should, in consultation with BAs, establish operating protocols with large loads that document expected behavior of large loads when reconnecting or ramping back up to expected load levels.

Real-Time Operations and Coordination

Large loads can impact real-time operations in various ways that need to be coordinated. As described in *White Paper: Characteristics and Risks of Emerging Large Loads*, a reliability concern is that large loads are interconnecting to the

⁸ E. Choukse et al., *Power Stabilization for AI Training Datacenters*, arXiv.org, Aug. 21, 2025. <https://arxiv.org/html/2508.14318>

grid for the first time within the near-term horizon. To mitigate this, TOs should establish requirements as described in the following sections in their interconnection agreements prior to energizing for the first time to ensure reliable coordination between large loads and system operators. This coordination should be completed using similar processes for generators and/or transmission being energized and connected for the first time.

Voltage Control and Coordination for Point of Interconnection

TOs, TPs, and PCs should coordinate to establish allowable power factor ranges for the large load point of interconnection as well as establishing allowable reactive power and voltage controls (static, dynamic, etc.) for the large load.

Outage Coordination Process Updates Incorporating Load

For established large loads, the outage coordination reliability requirements are needed to ensure that near-term assessments and real-time operations are evaluated properly. This is consistent with IRO-017 to maintain an outage coordination process for generation and transmission. Importantly, TOs and DP should coordinate with large load entities to report forced and planned outages of large load facilities to relevant grid operators (TOPs, BAs, and RCs).

The requirements should include the following:

- TOs and DPs should establish the following set of requirements in their interconnection requirements for reliable coordination between large loads and system operators:
 - Require large loads to provide and validate the modeling data needed for the BA, RC and TOP to conduct their analyses and assessments, including providing any updates, in advance, to modeling data due to modifications or upgrades.
 - Coordination to ensure operational readiness prior to energization
 - Verification that required network upgrades are completed prior to the energization
- Establish operating protocols that include items, such as:
 - Establish communication protocols
 - Establish operational large load facility net demand ramp rate limits in interconnection agreements in consultation with the TPs, PCs, RCs, TOPs, and BAs
 - Establish post-Disturbance voltage and frequency recovery requirements in consultation with BAs, RCs, TPs, and PCs. This should include, at a minimum, the characteristics needed to limit the maximum amount of load that could be disconnected (e.g., through tripping or customer isolation) for a single, credible System Contingency.
 - Process(es) for coordinating planned outages consistent with IRO-017
 - Establish a process to reliably integrate large loads onto the system, including coordination with RCs, TOPs, and BAs prior to energizing large loads to ensure that the large loads are properly accounted for in operational modeling, forecasting, or other operational requirements.
 - Establish a process for the expected large load to be accounted for in the near-term horizon forecast for the near-term energy reliability assessments (TOP-003 and BAL-007)
- RCs should establish the following set of requirements for reliable coordination with large loads:
 - Establish operating protocols that include process(es) for coordinating planned outages consistent with IRO-017 for generators and transmission.
 - Establish interpersonal communication capabilities similar to generators in COM-001.

Analysis of Customer-Initiated Load Reductions in Real-Time Contingency Analysis

TOPs, BAs, and RCs should incorporate statically defined customer-initiated load reduction contingencies into real-time contingency analysis and in operations studies. This should only be used if accurate large load facility fault performance data is unavailable and the large load entity cannot provide a clear estimation.

Large Load Communication Requirements

Large Loads can impact the reliability of the BPS if their real-time operations are not aligned with the operational criteria defined by the RC, BA, and TOP. This includes ramp rate limits, real-time dispatchability, providing day-ahead schedules, and data sharing. Similar to a generator, unexpected operations by a large load may raise concerns about balancing reserves and maintaining system reliability within system operating limits.

As mentioned in *Assessment of Gaps in Existing Practices, Requirements and Reliability Standards for Emerging Large Loads*, large loads are currently not registered entities and are not obligated to establish real-time communication with utility operators. Through a combination of TO interconnection requirements and NERC registration processes, the following mitigations should be considered:

- **Establish Interpersonal Communication Capabilities:** Require large loads to develop and maintain interpersonal communication capabilities that are essential for maintaining system reliability. This would be similar to the requirements set forth in COM-001 for GOPs. By implementing this requirement, large loads would be better equipped to communicate effectively with system operators, reducing the potential for miscommunications during critical situations.
- **Training and Adherence to Operating Instructions:** large load operators should undergo training and adhere to Operating Instructions akin to the standards outlined in PER-005 and COM-002 for GOPs. This would ensure that large load entities are well-prepared to respond appropriately during emergencies, thereby enhancing operational efficiency and safety.
- **Response to Operating Instructions:** Require operators of large loads to respond promptly and effectively to Operating Instructions issued by RCs, BAs, and TOPs as outlined in TOP-001 and IRO-001. This would align their responsibilities with those of other registered entities, ensuring a coordinated response during critical events and improving overall system reliability.

Event Analysis Process Changes

During normal grid operations, it is expected that the BPS will experience faults leading to disturbances and protection system operations. After such an event, the TO will typically conduct a review to analyze how the system performed to ensure that the BPS operated within its design criteria. With the increasing deployment of large loads, this event analysis process should be updated.

TOPs should monitor real-time performance of large loads (e.g., ramping, off-ramping, disconnection) to ensure its acceptability after energization. Additionally, large loads, as part of interconnection requirements, should be required to provide all necessary information needed for PCs/TPs/BAs/TOPs/RCs to conduct post-event analyses and assessments.

TOs/TOPs should have a monitoring system in place to ensure that the most up-to-date data is provided to BAs and RCs and other interested parties. The following approach together with a validation process could help mitigate the risks associated with collecting data for event analysis:

- **Develop monitoring strategy for monitoring large loads and incorporate corresponding requirements in TO's interconnection requirements and/or TOP's operating agreement documents**
- **Identify monitoring locations and develop a monitoring system to support real-time situational awareness and quick event data retrieving, including power quality monitor data, relay fault data, sequence of events, and phasor measurement unit recordings**

- Develop tools/applications to identify abnormalities and root causes, trigger alarms and send notifications, display events, and create event reports
- Develop pre-energization and ongoing event validation processes/procedures
 - Pre-energization validation to ensure the TOP is capable of monitoring the large load as designed in its real-time control center before it is declared ready for operation per applicable requirements/agreement
 - Ongoing event validation to help assess control setting performance per the large load facility design criteria

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Chapter 5: Stability

As outlined in the first LLTF white paper, large loads can significantly impact the stability of the BPS. As stated in [Chapter 2](#), TPs and PCs should perform required steady-state and dynamic studies under FAC-002 using the best available models provided by the large loads to identify the impacts and coordinate with the large loads to mitigate any risks pertaining to the BPS.

Major reliability risks may be hidden when large loads are not accurately represented in these studies, leading to unforeseen stability issues, such as the unexpected widespread tripping of large loads for a fault in an area. As mentioned in [Chapter 1](#), TOs and DPs, in coordination with TPs and PCs, should create interconnection request requirements that establish a minimum set of data and milestones that must be achieved prior to signing an interconnection agreement. This minimum data requirement should allow for the large loads to be fully and properly studied per FAC-002.

Voltage and Frequency Ride-Through

Voltage-sensitive large loads (i.e., data centers) have demonstrated load reduction and disconnection when exposed to normally cleared transmission faults.^{9,10} Load reduction in small quantities may not adversely impact BPS reliability, but at certain thresholds (local and interconnection-wide dependent), load reduction can cause cascading outages on a time scale of cycles to seconds.^{11,12} Thus, studying customer-initiated load reductions is a crucial part of ensuring reliable BPS operations. Large loads have not demonstrated similar sensitivity to frequency excursions.

In order to properly study the ride-through capabilities of large loads, large load entities should provide frequency and voltage trip settings to the TO, TP, and PC. TOs should update their interconnection agreements to require that this data be provided. For existing facilities, TOs should obtain voltage and frequency ride-through settings from large load facility owners that are verified via documentation from the OEM-provided operation manual of the equipment and strongly consider possible operational procedure changes and firmware upgrades, among other things, to ensure that the existing facilities can follow the ride-through criteria as closely as possible. If the fault performance data is not available for an as-built design of the large load facility, the TOs and TPs should require, and large load entities should provide, assumed trip settings.

TOs, TPs, PCs, BAs, and RCs should coordinate to determine local and system-wide thresholds for the magnitude of rapid demand change their area can withstand without triggering frequency or voltage stability limits. They should use the results of these studies to inform the need for establishing minimum voltage and frequency ride-through requirements (including rate-of-change of frequency (RoCoF) requirements), phase-jump considerations, or transmission mitigations to reduce system risk specific to their region in the interconnection requirements. TOs, in consultation with BAs, RCs, and TOPs, should establish these requirements for large loads considering the similar requirements for generating resources (e.g., PRC-019, PRC-024 and PRC-029). Ride-through requirements should be established on a per-interconnection basis to ensure that system-wide needs are met while avoiding mismatched protection settings. Additionally, the ride-through may-not-trip zone should be coordinated with UFLS trip settings.

Wherever a ride-through criteria cannot be established and TPs determine that an uncontrolled load disconnection poses a risk to system stability, the TP shall take actions to protect the security of the BPS. This could include imposing

⁹ *Incident Review Considering Simultaneous Voltage-Sensitive Load Reductions* -

https://www.nerc.com/pa/rrm/ea/Documents/Incident_Review_Large_Load_Loss.pdf

¹⁰ A. Springer, *Large Loads in ERCOT – Observations and Risks to Reliability*, presented at the Large Loads Task Force Kickoff Meeting, Oct. 2024, pp. 20–54. Available: https://www.nerc.com/comm/RSTC/LLTF/LLTF_Kickoff_Presentations.pdf

¹¹ ERCOT, *Load Loss Threshold Analysis*, presented at the ERCOT Large Load Working Group (LLWG) Meeting, May 2025. [Online]. Available: https://www.ercot.com/files/docs/2025/05/19/Large_Load_Loss_Analysis_051625_LLWG.pptx

¹² ERCOT, *Large Electronic Load (LEL) Voltage Ride-Through Overview*, presented at the ERCOT Large Load Workshop, June 2025, pp. 1–18. Available: https://www.ercot.com/files/docs/2025/06/13/COMBINED_June-2025-Large-Load-Workshop.pptx

consumption limits on large loads, installing dynamic VAR devices that can either prevent voltage sags from affecting the facility or prevent over-voltages resulting from sudden loss of the large load facility, or making other system changes as needed, such as having the BAs procure fast downward responsive resources that can manage the system overfrequency resulting from sudden load disconnection.

Large load entities should design their facility, including equipment protection and control settings, to reasonably accommodate the ride-through requirements in the interconnection requirements and coordinate with TPs and PCs on voltage and frequency ride-through settings.

While establishing limits for the rate and magnitude of large load facility net demand fluctuations, generator overfrequency limits should be considered to avoid cascading generator tripping. PRC-024 requires GOs to set their generator frequency protective relaying such that the relaying scheme does not trip the generator within the “no trip zone” of PRC-024 Attachment 1 with some exceptions. Attachment 1 requires continuous operation if the system frequency is less than 60.5 Hz for the Eastern Interconnection and 60.6 Hz for all other Interconnections. However, it is believed that not all generators can withstand a frequency rise of this magnitude and generator components will experience damage at this level. GOs should assess the capability of their generators to withstand the upper frequency limits and should inform BAs, TOs, and TPs of their equipment limits so they can be considered in large load ride-through analysis as required by R3 of PRC-024.

TOs, in coordination with TPs, PCs, RCs, and BAs, should create requirements for large loads connected through the same point of interconnection with generators to coordinate on-fault and post-fault performance. Subsequently, GOs, large load entities, and TPs and PCs should coordinate on-fault and post-fault performance between co-located generators and large load facilities (connected through the same point of interconnection).

Voltage Stability

Voltage instability caused by large loads can result in low- or high-voltage conditions, increased reactive power demand, and potential voltage collapse. To mitigate these risks, a structured approach across the planning, interconnection, and operational domains should be adopted. TOs should require large load entities to submit voltage ride-through curves, reactive power capabilities, and uninterruptible power supply (UPS)/tap-changer control settings as part of the interconnection package and ongoing data updates. This will facilitate the appropriate simulation of voltage recovery behavior under disturbances.

TPs and PCs should incorporate positive-sequence phasor-domain (PSPD) and EMT (if appropriate) dynamic models that adequately represent voltage-sensitive protections and controls. This allows evaluation of short- and long-term voltage recovery and dynamic reactive requirements. For large load facilities, utilities should consider their net load shapes (in case they have BTM generation) and the use of on-load tap changers (OLTC) at large load facilities to assess their impact on long-term voltage stability. Long-term voltage stability analyses are performed using the classic P-V/Q-V analyses in power flow domain. For large load facilities, one should consider their net load shape to identify the loading levels during peak load conditions where voltage stability issues generally appear. Studies should also consider if any BTM or collocated generation is available to mitigate the voltage issues. Where necessary, TPs should assess the potential need for dynamic reactive support such as static VAR devices or synchronous condensers.

TPs and PCs should run contingency-based voltage assessments that include realistic load trip/reconnect behaviors, evaluating system VAR margins and identifying locations where rapid voltage recovery is not possible. TOs should coordinate with RCs and TOPs to define restoration sequences that avoid simultaneous reconnection of large blocks of load. Real-time alarms should be configured to detect declining voltage stability margins (e.g., RoCoV, low-Q reserve). Voltage-related event analysis should include comparison of observed large load behavior to modeled expectations, with models adjusted accordingly. Post-event findings should inform ongoing model improvements and settings refinements.

TPs and PCs should also assess voltage stability, including under/over-voltage performance, the potential need for under/over-voltage protection schemes, and the Most Severe Load Contingency (MSLC) for each load cluster. Due to the local nature of the voltage issues, the assessment should be performed for each clustered load of concern. This differs from the frequency assessments, where a single MSLC can be applied at system level.

These voltage stability mitigation practices should be reflected in the interconnection agreement and assumptions should be updated periodically with operational data and commissioning results.

Frequency Stability (Overfrequency)

Rapid changes in real power demand, including large load reductions or ramping, may introduce frequency disturbances and instability. The most consequential and credible mechanism observed to date is aggregate customer-initiated load reduction (CILR) following transmission faults. These events create a nearly instantaneous imbalance where generation exceeds demand by the amount of CILR. The large imbalance of generation and load drives system frequency upward at a rate determined by the magnitude of load reduction, the interconnection's online inertia, and primary frequency response. The largest RoCoF would be observed when interconnection inertia and available primary frequency response are at minimums. The three primary variables involved in this problem are largest aggregate CILR, online inertia, and frequency-responsive reserves.

The first variable is the largest aggregate CILR which is dependent on the location of the fault and the amount of voltage-sensitive load that is electrically close to the fault. To determine the amount of the largest aggregate CILR, TOs should collect, and large load entities should provide, detailed information on voltage conditions that will cause large load facilities on their system to reduce output or disconnect from the grid. Additionally, TOs should collect supporting information such as the UPS transfer logic and backup generator synchronization which should be modeled and validated in dynamic simulation studies (e.g., via PSPD or EMT tools). This should include delay logic, reconnection criteria, and voltage/frequency tolerance ranges. TOs should share the collected dynamic modeling information with their TP and PC. This dynamic model should go through model verification and model validation processes, as mentioned in [Chapter 1](#).

On an ongoing basis, TPs should use large load facility dynamic model information to assess and report to their BA the largest credible CILR due to a worst-case fault on their system. TPs should also plan the system in such a way that the largest credible load reduction is limited to a level that is acceptable for the BA and PC regions they operate in.

The second variable of frequency stability for CILR events is the amount of on-line inertia, which is variable and depends on what synchronous generators are on-line. CILR events that happen when inertia is lowest will have the highest RoCoF. BAs and PCs should account for the expected amount of on-line inertia when procuring Operating Reserves that will adequately ensure reliability given the worst case CILR event. The third variable of frequency stability for CILR events is the amount of on-line frequency-responsive reserves. These reserves are comprised of on-line generation with legroom that has been tested and verified to be capable of providing droop as described in the *Primary Frequency Control Reliability Guideline*.¹³

To mitigate the risk posed to frequency stability, BAs and PCs in each interconnection should analyze and study the worst-case scenario(s) where the system is shown to have at- or near-minimum inertia and primary frequency-responsive reserves. The analysis should include the worst-case fault(s), analyze system frequency stability, and identify any generators that may trip off-line.

Through a combination of the aforementioned risk mitigations to frequency stability risks, interconnection frequency should remain stable and not infringe on any overfrequency limits that may impact reliable BPS operations (cascading

¹³ "Primary Frequency Control Reliability Guideline," NERC. Available:

https://www.nerc.com/comm/RSTC_Reliability_Guidelines/Primary_Frequency_Control_Reliability_Guideline_v_4.0_Clean_V3%20-%20ODR.pdf

generator trips). If the system frequency is shown to be stable in simulation and in practice for worst-case or near worst-case scenarios, the risk is deemed to be mitigated.

Oscillation Mitigation

Certain large loads with variable and cyclical profiles, such as AI data centers, can introduce forced oscillations into the BPS. These loads often operate cyclically. If the frequency of load cycling is in the electromechanical range (i.e., 0.1–1.5Hz), there is a potential for interacting with natural low frequencies and causing widespread forced oscillations. Such oscillations pose significant risks to the BPS and should be studied before these loads are energized. For oscillations at higher frequencies, risks of subsynchronous control or torsional interactions exist.

Since large loads are currently not registered entities, TOs should consider incorporating monitoring and mitigation provisions in interconnection agreements to address potential oscillatory behavior. TOs and RCs should revise their oscillation monitoring/mitigation platforms and procedures to account for potential load-induced oscillations. If wide-area oscillations are observed, coordinated actions by multiple RCs may be needed to address the reliability concern. RCs should incorporate large load-specific actions in their coordination plans. Adverse reliability impacts from oscillations that occur may be mitigated through coordinated use of supplemental controls, such as nearby flexible alternating current transmission system (FACTS) devices or generator-based stabilizers, and provisions for attenuating forced oscillations via technologies like battery energy storage systems (BESS). Additionally, TOs, TPs, PCs, and RCs should collaborate to establish requirements for large loads to avoid exciting specific modes of the local system and interconnection.

Cyclical power demand from large load facilities can significantly impact the stability of electric grids. In order to properly study the effects of cyclical demand, the data mentioned in [Chapter 1](#) should be collected. This fluctuating demand has the potential to excite torsional modes in nearby generators, leading to mechanical stress and potential damage. Additionally, it can interfere with the natural modes of other equipment, such as the controls of IBRs. Additionally, large load facilities must be assessed for their potential to contribute to oscillatory instability. This can occur if device controls in large load facilities interact unfavorably with the network impedance or other nearby equipment, potentially leading to sustained power oscillations.

As these system interactions can cause significant risk to the local transmission system, generators, and the entire interconnection, the TO should also incorporate monitoring and protective equipment requirements for large loads as required by the TOP and RC. TPs, PCs, and large load entities should consider following recommendations to identify mitigation (e.g., E-STATCOM, BESS) and reduce potential risks associated with oscillation.

To support both proactive and post-event mitigation, the following practices are recommended:

- **Detection and Mitigation**
 - Data from these monitoring devices should be incorporated into existing oscillation detection and source identification platforms that the TOs and RCs are using. PMU measurements are best suited for detecting/analyzing oscillations below 10 Hz, while the accurate characterization of higher-frequency oscillation requires point-on-wave measurements.
 - In the operations time frame, if the TO/TOP/RC detects low-damping conditions for a natural mode, they can consider establishing mechanisms to request large loads to defer oscillatory behavior as long as the low-damping conditions persist.
- **Interconnection Requirements:**
 - TPs and PCs should provide expected system strength ranges at the POI for equipment validation.
 - TPs and PCs should establish oscillation damping metrics and thresholds that loads must meet under interconnection requirements, in accordance with TPL-001 standards. They should also establish limits

(amplitude and frequency) on oscillatory demand from large load facilities depending on the risk of sub synchronous resonance (SSR) or exciting any interarea modes. Industry-vetted methodologies for establishing these limits do not currently exist. An empirical rule that may be followed is avoiding natural frequencies excitable from a large load location. These frequencies and high-risk locations may be identified through planning studies.¹⁴

- TPs and PCs should identify weak grid areas or locations prone to inter-area oscillations and restrict load interconnections accordingly. To support this requirement, the definition of short-circuit ratio likely must be revisited. For a load co-located with a generator, there is no clear guidance on whether the SCR should be calculated based on load rating, generator rating, or net load rating.
- **Monitoring Requirements:**
 - High-resolution monitoring as detailed in [Chapter 4](#) can support real-time oscillation detection and post-disturbance verification of load behavior. Additionally, grid operators can identify load facilities that are contributing to oscillations, informing mitigation actions that could include isolating the source load. PMU measurements are best suited for detecting/analyzing oscillations below 10 Hz, while the accurate characterization of higher-frequency oscillation requires POW measurements.
 - Data from these monitoring devices should be incorporated into existing oscillation detection and source identification platforms that the TOs and RCs are using.
 - In the operations timeframe, if the TO/TOP/RC detects low-damping conditions for a natural mode, they should consider establishing mechanisms to request large loads to defer oscillatory behavior as long as the low-damping conditions persist.
- **Event Analysis and Model Validation:**
 - If a TO or TOP determines that a large load facility is contributing to system oscillations, the TP and PC should conduct post-event analysis using as-built models to assess damping behavior and determine if any equipment will need to be installed to mitigate the oscillation.
 - Time-synchronized PMU, POW, and SCADA data should be analyzed to compare observed response with simulation expectations. Model adjustments should follow as needed.
 - If repeated oscillatory behavior is observed, the TOs and RCs may define protective relaying or disconnect logic to safeguard grid stability.

Angular Stability

Angular instability may arise in systems with significant CILR events or rapid real power demand fluctuations, particularly during faults in weak or constrained areas. While this risk is often less emphasized than voltage or frequency stability, under certain network conditions, it may result in inter-area angle swings or loss of synchronism. See [Figure 5.1](#) demonstrating loss of synchronism in a PSPD simulation.

Recent industry experience highlights a specific angular stability concern involving large loads co-located with generation in areas with weak transmission connectivity. In such scenarios, the sudden loss of load can leave nearby generators without sufficient transmission capacity to evacuate power to the broader grid, leading to large angle swings or potential loss of synchronism. This condition is particularly pronounced when both the generation and load are part of a combined facility with limited ties to the rest of the system.

¹⁴ S Biswas, A C Varghese, K Chatterjee, et al., *Evaluating the Risk to Bulk Power System Reliability from Large Load Induced Oscillations*. TechRxiv. August 26, 2025.

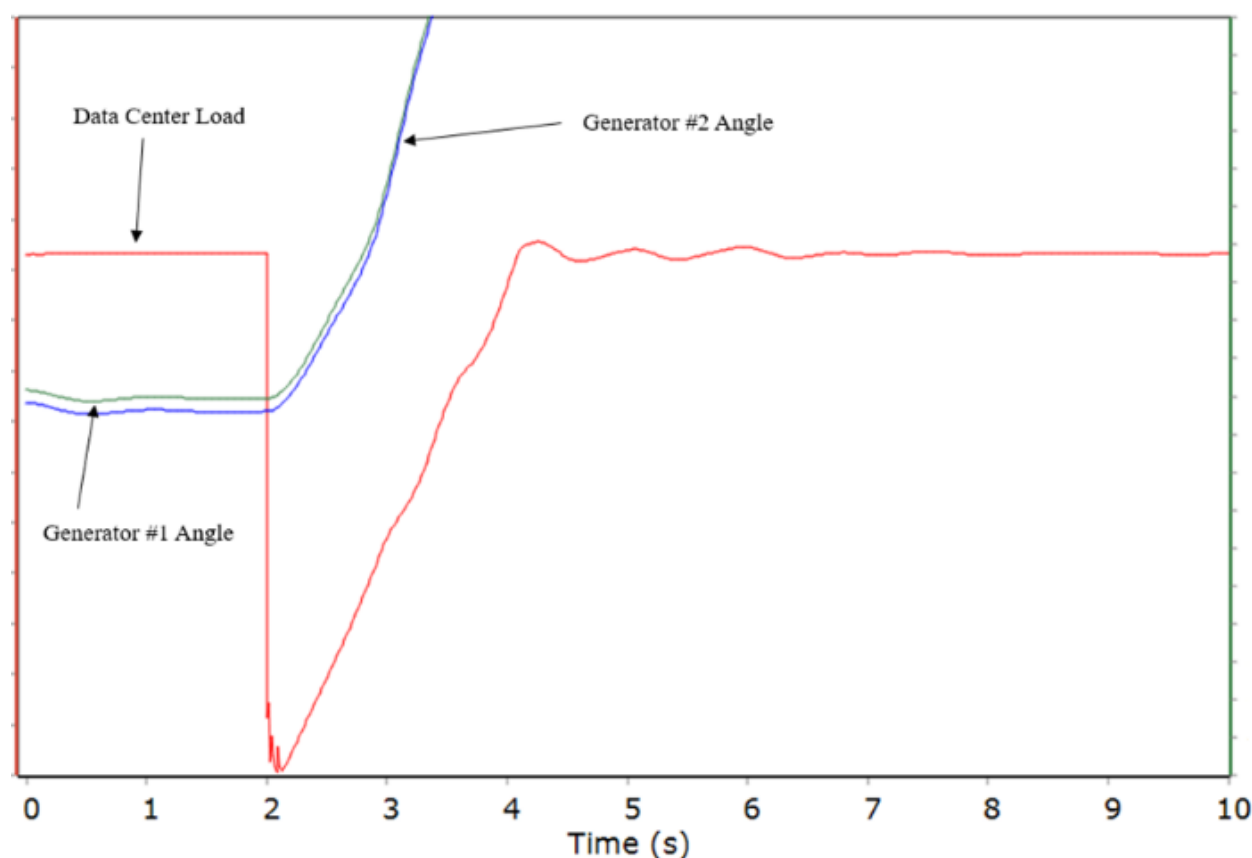


Figure 5.1: Multiple Generators Lose Synchronism Following a Data Center Load Trip and Subsequent Rapid Recovery

TOs and PCs should consider angular stability risks when evaluating large loads in areas with long radial lines or high impedance corridors or during scenarios involving delayed fault clearing. Interconnection studies should include angle margin assessments using time-domain simulations. EMT or RMS simulation models should represent the fast dynamics of load components, including converter controls, UPS synchronization, and backup generation transitions. This is necessary to assess system response to disturbances. For large load centers, planners should study the impact of large load trip or recovery during N-1 and N-1-1 contingencies, especially in low-inertia regions. This helps determine the risk of angular separation or uncontrolled power swings.

TOs should request oscillography data from staged fault tests and commissioning events to validate angular response assumptions. PMUs installed at the point of interconnection of large load facilities should monitor phase angles, RoCoF, and modal interactions during events.

If angular instability is identified as a risk, mitigation may include synchronous condensers, wide-area damping control schemes, or redesign of restoration sequences. Remedial Action Schemes (RAS) may also be considered. Event analysis post-fault should examine angle trajectories and coherency groups to determine if large load behavior was a contributing factor to any loss of synchronism or instability. TOs and PCs should utilize real-time monitoring to monitor for angular stability risks. TOs, BAs, and RCs should also determine if it is appropriate to install protective equipment to automatically disconnect the large loads to protect the system from oscillations. TOs and TPs should update modeling assumptions accordingly based on system changes and identified control or stability issues.

Chapter 6: Power Quality

Emerging large loads often contain significant quantities of electronic power supplies, adjustable speed drives, and rectifiers, which can generate harmonic distortion and voltage fluctuations in the BPS. TPs and PCs should perform required studies under FAC-002 using the as-planned models provided by the large loads to identify power quality impacts and ensure that they are within the acceptable levels or if any mitigation measures will be necessary. Studies should be conducted under the diverse operating modes of large loads while also considering various potential load equipment configurations.

Harmonics

When considering harmonics, power quality standards (e.g., IEEE Std 519-2022) already exist that specify harmonic limits with which all large loads should be required to comply. Since existing power quality standards are not mandatory, TOs or DPs should include harmonic limits in their interconnection requirements for large loads, as well as any RC, BA, or TOP requirements specific to their footprint into their interconnection requirements under FAC-001.

Using the data provided by the large load, the TP and PC should use as-planned models to ensure that individual and overall current harmonics injections into the interconnecting substation are within the limits. During the Power Quality studies, harmonic evaluation studies may be performed by the TP and/or PC to do the following:

- Determine the impact of large load-produced harmonic current injections on electrically close synchronous generators to ensure that the levels are within the specified machine ratings (i.e., permissible continuous equivalent negative-sequence current capability).
- Determine the potential impact of large load-produced harmonic current injections on existing harmonic filter banks (i.e., impact of harmonics on individual filter component ratings).
- Determine the impact of large load-produced harmonic current injections on the overall voltage distortion at the interconnecting substation.
- Determine the harmonic spectrum of the large load under various grid conditions, including low short-circuit ratio (SCR) scenarios.
- Determine if mitigation measures will be necessary to meet the machine or harmonic filter ratings. In cases where technical solutions are insufficient for meeting harmonic limits, alternative approaches like utility-scale battery storage may be needed.

To support the analyses above, load owners should provide the current harmonic spectra to the TP enabling estimation of the individual harmonic current injections (up to 50th harmonic order) at the interconnecting substation.

Voltage Fluctuations and Flicker

Flicker analysis may be performed based on the expected load profile showing variations in active and reactive power that capture any cyclic changes. The analysis will estimate the flicker contribution of the large loads at the interconnecting substation and should be compared against the permissible limits in the TO's interconnection requirements related to power quality. IEEE Std. 1453-2022 has well-defined requirements for flicker that TOs and DPs should consider adopting into their interconnection requirements. This analysis may identify the need for mitigation measures to bring flicker contribution of the large loads to acceptable levels.

To support the analyses above, load owners should provide active and reactive power profiles that are detailed enough to allow for both short- and long-term flicker to be estimated.

Managing Facility Design Changes that Impact Power Quality

The harmonic emissions and voltage fluctuations associated with a particular large load may vary significantly with design decisions that may not be made until the interconnection process is well underway (e.g., the selection of specific vendors for electronic devices or determination of the site's mix of computational processes). In such cases, the load facility design changes may impact the load facilities' compliance with power quality requirements. TOs and DPs should have a clear process on how and when large load facility design changes are communicated to them by the large load entity. In these situations, the large load entity may experience cost increases and schedule delays if TO and DP processes are unclear on how design changes are communicated and handled.

Even after a large load is put into service, significant changes in its load profile (e.g., switching from traditional cloud services to large-language model training) could result in power quality violations that may not have been studied in earlier studies. It may then become necessary to specify and install additional equipment (e.g., harmonic filters or active power conditioning devices) to mitigate the violation, incurring additional costs and potentially altering the interconnection timeline.

TOs and DPs should have clear policies in place to handle such scenarios along with the appropriate measures that may be taken to ensure system reliability (e.g., operational restrictions or other mitigations). Additionally, based on the severity of the violation, timelines for remediation could be determined without immediately impacting the operational status of the project (e.g., a minor violation may be permitted to continue without operational downtime for the project so long as it is resolved within a reasonable amount of time).

TOs and DPs should have policies for addressing unanticipated power quality impacts that are not covered by limits set forth in interconnection agreements. For example, it is possible that demand fluctuations produced by a large load will cause the lights of other nearby customers to flicker even if the load owner is subjected to and compliant with relatively strict interharmonic limits. There is also the possibility of unforeseen interactions between equipment at multiple large load facilities that are electrically close to one another. TOs and DPs should have policies in place that enable cooperation with load owners to resolve these issues.

Power Quality Monitoring and Analysis

TOs should install high-resolution data monitoring equipment at the point of interconnection substation to validate the accuracy of power quality data (e.g., load profile), compliance with power quality performance limits, and whether the large load is participating in system power quality issues. In the event of system-level power quality impacts, TPs and PCs should use the validated as-left load models to perform power quality studies to mitigate any identified deficiencies and propose solutions.

Chapter 7: Physical and Cyber Security

Introduction

This chapter provides recommendations for NERC registered entities and large load entities to systematically integrate physical and cyber security considerations into the planning, design, and integration processes for new or existing large industrial loads. The objective is to ensure that the integration of such large load facilities maintains or enhances the reliability, resilience, and security of the electric grid, consistent with NERC Reliability Standards and good utility practice. This chapter is informed by the initial work of the NERC LLTF as documented in its white papers, *Characteristics and Risks of Emerging Large Loads* and *Assessment of Gaps in Existing Practices, Requirements and Reliability Standards for Emerging Large Loads*. It emphasizes proactive risk mitigation for both physical and cyber threats, recognizing the potential impact of large load facilities on regional and interconnected system stability.

Background

The increasing demand for large loads introduces unique physical and cyber security challenges and potential vulnerabilities to the BES. The LLTF has identified that these large loads, which often include complex microgrids and BTM generation, have unique characteristics, such as rapid growth and shorter interconnection timelines. The LLTF's initial *White Paper: Characteristics and Risks of Emerging Large Loads* highlights a range of risks posed by large loads, including those related to dynamic modeling, resource adequacy, and stability. While the paper's initial risk prioritization categorized security risks as "Medium" for further assessment, this guideline provides risk mitigations for addressing both physical and cyber gaps as they relate to these entities and their potential to impact BPS reliability. Until such time where large load entities are NERC registered entities, the existing NERC registered entities that interact with the large loads must bear the responsibility for ensuring BPS reliability and compliance with NERC security mandates. These security mandates are consistent with Critical Infrastructure Protection (CIP) and Operations and Planning (O&P) Reliability Standards, including physical security, cyber security, and operational requirements such as EOP-004 event reporting.

Principles for Physical and Cyber Security Integration

Holistic Risk Assessment

NERC registered entities and large load entities should conduct comprehensive, forward-looking risk assessments that integrate physical and cyber threat vectors impacting both the large load facility itself and its point of interconnection with the BES. This assessment should consider the potential for cascading impacts on the wider grid. The assessment should also account for the unique operational characteristics and potential vulnerabilities of diverse large loads, including data centers and industrial facilities with BTM power generation and storage resources, as well as demand-side management capabilities, recognizing that current NERC CIP Reliability Standards may not be sufficient in granularity for these unique aspects. The results of this assessment should be used to inform the development of specific security requirements to be included in interconnection agreements and tariffs.

Security-by-Design

NERC Registered Entities and large load entities should advocate for and require the incorporation of physical and cyber security controls as foundational elements during the design phase of new large load interconnections rather than as retrofitted additions. This approach should address new cyber attack vectors introduced by the growing adoption of internet-connected process control networks (PCN), operations management networks (OMN), and industrial control systems (ICS) and the potential for these loads to participate in energy markets as aggregated distributed energy resources. These foundational controls should be formalized in the interconnection agreements and tariffs governing the relationship.

Operational Interdependency Recognition

NERC registered entities and large load asset owners must acknowledge and plan for the operational interdependencies between the large load's internal systems (e.g., industrial control systems, microgrid controls) and the utility's BES control systems, ensuring secure and reliable data exchange and control capabilities. The specific requirements for secure and reliable data exchange along with proper management of BCSI should be a core component of the interconnection agreements.

Collaboration and Information Sharing

NERC registered entities and large load asset owners should foster effective communication and collaboration from the earliest stages of a project. This collaboration is crucial for identifying, assessing, and effectively mitigating evolving security challenges and may necessitate updates to existing NERC Reliability Standards and the development of new best practices. The interconnection agreement and applicable tariffs serve as the primary legal and contractual mechanisms to formally define the terms of this collaboration and information sharing.

Physical Security Considerations for Large Load Interconnection

Site Selection and Layout Review (CIP-014 Alignment)

- **Purpose:** To identify and mitigate physical security risks associated with the location and configuration of large load facilities and their immediate interconnection infrastructure.
- **Guideline:** NERC registered entities and large load asset owners should perform an initial assessment of the potential criticality of the proposed interconnection substation and associated transmission elements if subjected to physical attack (aligned with CIP-014 principles). This assessment should recognize the gap in mandated depth or periodicity of threat assessments for some large load categories, as CIP-014 is not applicable to them. The physical environment, accessibility, and visibility of the proposed site should be analyzed, and design features that inherently enhance physical security should be recommended. This includes advising on robust perimeter security for critical components. These site-specific physical security requirements must be documented in the interconnection agreement.

Equipment Protection and Resilience

- **Purpose:** To ensure that critical equipment supporting the large load interconnection is adequately protected against physical threats and can withstand or recover from attacks.
- **Guideline:** Recommend hardening measures for critical control houses, communication infrastructure, and vulnerable equipment for large load assets. This includes physical security for control cabinets or converter stations for devices like STATCOM/FACTS/BESS/HVdc converters as well as adequate physical hardening of substations where these devices are installed. Opportunities for redundant physical pathways or diverse equipment routing should be explored to minimize single points of failure. Both NERC registered entities and large load asset owners should also coordinate on strategies for rapid replacement of critical long lead-time equipment. Specific requirements for equipment protection, redundancy, and spare equipment strategies should be articulated in the interconnection agreement.

Table 7.1: Enhanced Security Matrix—Sector Frameworks, NERC Entity Mandates, and Large Load Gaps

Sector	Security Frameworks	NERC Registered Entities and CIP Mandates	Large Load Gaps	LLTF Guideline Implications
Hyperscale Data Centers	Uptime Tier System, NIST CSF	TO/TOP: CIP-014 (physical security), CIP-006 (access control) RC/BA: CIP-002 (BCS identification), CIP-008 (incident response) TP/PC: TPL-001, MOD-025/026/027 (modeling and planning)	- Not subject to CIP unless BES-connected - No mandate for cyber-physical integration - Often distribution-level	- Require cyber-physical controls for Tier III/IV - Enforce observability and data-sharing protocols
Hydrogen Fuel Production	IEC 62443, fragmented DOE/EPA/OSHA oversight	TO/TP/PC: CIP-005 (electronic perimeter), CIP-007 (system security) RC: CIP-010 (configuration management), CIP-011 (data protection)	- No unified cyber security standard - Often modular and unregistered - Legacy ICS vulnerabilities	- Promote secure ICS architecture - Push for sector-specific cybersecurity baselines
Large Industrial Loads	IEC 62443, ISO/IEC 27001, CMMC, NIST CSF	TO/TP/PC: CIP-013 (supply chain risk), CIP-004 (personnel and training) BA/RC: CIP-009 (recovery plans), CIP-012 (data comm. protection) TOP: CIP-003 (security management controls)	- Not NERC registered entities - No personnel screening mandates - Insider threat risk unaddressed	- Recommend personnel vetting - Harmonize cross-framework security practices

Detection, Surveillance, and Response Integration

- **Purpose:** To ensure that physical security measures at the large load's interconnection point are integrated into broader utility security operations.
- **Guideline:** NERC registered entities and large load asset owners should implement and integrate surveillance (CCTV), intrusion detection systems, and access control systems. This should address the lack of physical security measures for critical OT components like RTUs, PMUs, IEDs, STATCOM control houses, BESS containers, HVdc converter stations, and communication nodes. Joint discussions and planning between the large load entity, the interconnecting utility, and local law enforcement/first responders should be facilitated for unified response to physical security incidents. The specific requirements for security technology integration and joint response planning should be outlined in the interconnection agreement.

Cyber Security Considerations for Large Load Interconnections

Cyber System Identification and Classification (CIP-002, CIP-003 Alignment)

- **Purpose:** To identify and appropriately classify BES Cyber Systems (BCS) and associated communication networks related to the large load interconnection.
- **Guideline:** NERC registered entities and large load asset owners should clearly define the boundaries of any BCS within the large load facility that directly impact the reliable operation of the BES. It must be acknowledged that large loads are not NERC registered entities, and thus devices, systems, tools, and protocols associated with their BTM operation and maintenance of grid-connected electrical infrastructure

are not subject to NERC mandates, creating an inherent risk. They should work together to determine the appropriate impact rating (High, Medium, Low) for these identified BCS. The final classification and identification of these systems should be a required annex to the interconnection agreement.

Secure Network Design and Architecture (CIP-005, CIP-007 Alignment)

- **Purpose:** To ensure that the network architecture supporting the large load interconnection is designed to prevent, detect, and respond to cyber incidents.
- **Guideline:** NERC registered entities and large load asset owners should mandate robust network segmentation between the large load's OT and IT networks. They should require the use of encrypted and authenticated communication protocols for all data exchange. They should also establish strict controls and secure methods for any remote access to critical systems at the large load facility that could impact the BES. Specific technical security requirements and standards must be formalized in the interconnection agreement and referenced in applicable tariffs.

Supply Chain Risk Management (CIP-013 Alignment)

- **Purpose:** To mitigate cyber security risks introduced through the supply chain of equipment and software utilized in the large load interconnection.
- **Guideline:** NERC registered entities should advise large load asset owners on conducting cyber security due diligence on vendors providing control systems, network devices, and critical software for their interconnection. They should also encourage secure configuration baselines for all new hardware and software to minimize default vulnerabilities. The interconnection agreement can require attestations from the large load entity regarding its supply chain due diligence.

Vulnerability Management and Patching (CIP-007 Alignment)

- **Purpose:** To ensure ongoing management of vulnerabilities in systems related to the large load interconnection.
- **Guideline:** NERC registered entities and large load asset owners should recommend regular vulnerability assessments and penetration testing for critical systems within the large load's operational environment that interface with the BES. They should also establish protocols for coordinating patching and security updates between the large load entity and the interconnecting utility. The interconnection agreement should outline the frequency of these assessments and the protocols for coordinated patching.

Incident Response and Recovery Planning (CIP-008 Alignment)

- **Purpose:** To ensure the capability to detect, respond to, and recover from cyber security incidents impacting the large load interconnection.
- **Guideline:** NERC registered entities and large load asset owners should facilitate the development of joint cyber security incident response plans. They should also emphasize robust data backup and system restoration capabilities for critical systems at the large load facility that impact BES reliability. The interconnection agreement must include a jointly developed and tested incident response plan with clear communication protocols, roles, and responsibilities. Large load asset owners could encourage staff participation in E-ISAC GridEx exercises.¹⁵

¹⁵ E-ISAC's GridEx exercises are comprehensive, large-scale, biennial events that simulate coordinated cyber and physical attacks on North America's electric grid to test and improve crisis response and resilience.

Documentation, Review and Continuous Improvement

Documentation Requirements: All analyses, recommendations, and agreements stemming from the physical and cyber security assessments shall be thoroughly documented. This documentation, including the final interconnection agreement, serves as the authoritative record of security requirements.

Review and Update Cycle: This guideline and its application should be reviewed periodically to ensure continued relevance and effectiveness. Any updates to the security posture should be formally amended into the interconnection agreement.

Training and Awareness: NERC, along with large load trade organizations, should promote training and awareness programs for both investor owned utilities (IOU), grid operators, privately held transmission/distribution firms, and large load personnel on the importance of physical and cyber security. This should address the critical gap regarding personnel security and training for large loads, as NERC Reliability Standards like PER-003-2, PER-005-2, PER-006-1, and CIP-004 are not currently applicable to large loads. The lack of auditable requirements for limiting access and control of large loads and the lack of requirements for training of operators of large loads is a significant gap in NERC Reliability Standards. The interconnection agreement can be a vehicle to require personnel training and vetting for the large load asset owner's staff.

Chapter 8: Resilience, System Restoration, and Load Shedding

Introduction

Resilience is the ability to withstand and reduce the magnitude and/or duration of disruptive events. Resilient infrastructure facilitates the ability to anticipate, absorb, adapt to, and/or rapidly recover from such an event. Typical BES resilience infrastructure facilities include manual load shedding, automatic load shedding, and system restoration.

Large loads, with their high power ratings and dynamic characteristics, can introduce challenges to the BES resilience. This chapter analyzes reliability risks posed by large loads, identifies key gaps, and proposes mitigation measures to enhance BES resilience.

Load Shedding

Manual Load Shedding

NERC EOP-011 requires BAs and TOPs to have the capability and provisions to implement manual load shedding (MLS) to mitigate emergencies. Additionally, it mandates provisions to minimize the overlap among circuits designated for MLS, UVLS, and UFLS. In areas with a high penetration of large loads, avoiding overlap among these programs may be difficult if large loads are excluded from participation. Therefore, TOs should consult with BAs and TOPs to determine whether including large loads in MLS plans is appropriate.

Given that the power consumption of large loads is variable, or constant but not following the general system load profile, shedding them based on the UFLS prediction could result in the TO either failing to meet the load-shedding obligation or shedding too much load, leading to overfrequency and/or overvoltage situations. The following recommendations aim to address these issues:

- **Monitor Actual Load Consumption:** Real-time monitoring, such as SCADA or telemetry, is critical for TOPs to accurately control the amount of load shed. Large loads participating in MLS should be equipped with such monitoring systems.
- **Partial Load Shedding:** Segmenting a large load into controllable blocks allows for more precise load shedding while maintaining service to other parts. TOs and RCs should develop guidelines and standards for partial load shedding of large loads in coordination with large load entities.
- **Incorporate Data Center Flexibility:** Data centers have inherent flexibility to rapidly change their demand by rescheduling or shifting machine learning workloads and non-urgent compute tasks to support the grid.¹⁶ Agreements and protocols can be set up between TOs and large load entities on integrating these flexibilities to support grid when necessary.
- **Coordination with Load Owners:** TOs/TOPs and large load entities should establish communication pathways to coordinate expected load shedding needs.
- **Regular Obligation Review:** RCs and TOs should increase the frequency of reviewing load shedding obligations to keep pace with large load growth.

Automatic Load Shedding

NERC PRC-006 defines design and documentation requirements for automatic UFLS programs. The aim of these programs is to balance system load with available generation sources in the event of a system-wide incident that causes a shortage of available generation to meet the connected load. Due to the size of large loads ranging from hundreds to thousands of megawatts, as well as the high penetration of multiple large loads in a given PC area, it may be necessary to include large loads as part of UFLS schemes to satisfy the PRC-006 performance criteria.

¹⁶ <https://blog.google/inside-google/infrastructure/how-were-making-data-centers-more-flexible-to-benefit-power-grids/>

Therefore, TOs should coordinate with PCs to determine if it is appropriate to include the large loads in their UFLS programs (e.g., TOs and PCs may require large loads to design facilities to have the capability to participate in automatic UFLS or in distribution management system (DMS) control.)

PRC-006 requires the frequency of each identified island to remain within the defined range for underfrequency conditions resulting from an imbalance of up to 25%. The departure of actual UFLS stage percentages from obligation percentages at any time could result in insufficient or excessive UFLS capability available from PRC-006 design basis of 25%.¹⁷ For a UFLS program originally established to include at least 25% of the total load, when one or more non-UFLS participating large load(s) are added to the total load, the original stationary UFLS percentage may not sustain and need to vary depending on if the large load is on-line or off-line and if the other loads are at peak or minimum demand. Keeping the original static UFLS percentage could cause insufficient or excessive UFLS and may result in the frequency excursion. The recommendations include the following:

- **Improved Observability:** TO, TOP, DP and other UFLS entities should monitor real-time changes in UFLS load percentage.
- **Set Load Variability Bounds:** PCs should define acceptable load variability ranges.
- **Assess Technologies to Provide Frequency Support:** Technologies such as energy storage can potentially support frequency and voltage dips during disturbances.
- **Coordinate Across Areas:** BAs should coordinate to synchronize UFLS staging between exporting and importing areas, which can reduce transmission stress.
- **Coordinate with Large Load Protection and Control Settings:** Large Loads are sensitive to system frequency/voltage changes and can automatically switch to internal backup generation (equivalent to large load shedding for the grid) when large frequency/voltage deviations are observed at their terminals. Thus, TOs and BAs should consider such behaviors in their UFLS settings and program design.

Similarly, NERC PRC-010 establishes an integrated and coordinated approach to the design, evaluation, and reliable operation of UVLS programs. In some cases, large loads may be of significant size or clustered in an area where inclusion of large loads as part of UVLS schemes is needed to satisfy the PRC-010 performance criteria. Therefore, TOs should coordinate with PCs to determine if it is appropriate to include the large loads in their UVLS programs.

Both PRC-006 and PRC-010 require each PC or TP to evaluate the effectiveness of each UFLS and UVLS program at least every five years. Given that large loads can be constructed in as quickly as 12 –18 months and due to rapid large load deployments, an assessment every five years may not be sufficient. PCs and TPs should dynamically adjust the UFLS and UVLS assessment frequency based on the load growth rate of their system or define a system load deployment threshold that triggers new UFLS or UVLS assessments if the threshold is passed. This will ensure UFLS and UVLS reliability in areas with fast load growth. If new assessments are not performed, then the BPS may collapse if UFLS or UVLS schemes are triggered but prove inadequate due to load growth.

Because large loads are not registered entities under NERC, if a TO determines that a large load is part of its MLS, UVLS, or UFLS schemes,¹⁸ the TO should require the large load to coordinate with the testing, maintenance, and design scheme (e.g., feeder breaker tripping) to support implementation.

¹⁷ Robert J O’Keefe, *Proposed PRC-006 SAR for UFLS Variability Monitoring and Stage Percent Bounds*, NERC LLTF, June 18, 2025. Available: https://www.nerc.com/comm/RSTC/LLTF/LLTF_June_Workshop_Presentations.pdf

¹⁸ There is a DP-UFLS registered entity according to NERC ROP Appendix 5B: A Distribution Provider entity shall be an Underfrequency Load Shedding (UFLS)-Only Distribution Provider if it is the responsible entity that owns, controls or operates UFLS Protection System(s) needed to implement a required UFLS program designed for the protection of the BES, but does not meet any of the other registration criteria for a Distribution Provider. A DP designated as ‘DP – UFLS Only’ will only be accountable for a sub-set of the NERC Reliability Standards as described in Section III.b.2 of NERC ROP Appendix 5B, Statement of Compliance Registry Criteria.

Additionally, due to the voltage sensitivity of some types of large loads and the potential for simultaneous disconnection during a single fault or disturbance due to collective ride-through behavior, TP and PCs should also assess frequency stability, including overfrequency performance, the need for overfrequency protection schemes, and the MSSC on the demand side. TPs should require large load entities to submit the ride-through capability data of the load and the test report including the as-built ride-through characteristics, which must be modeled and considered in the planning and operational analyses.

Dynamic Load Shed Monitoring and Allocation

If large loads are included in load shed plans such as MLS, UVLS, or UFLS, then TOs and DP should coordinate on a real-time basis if dynamic computation of load shed allocation is implemented. TOs and DP should develop communication protocols to automatically communicate load shed percentages in real time to protection systems in the field. TOs should assess the need for a wide-area protection scheme that dynamically sets and adjusts load shed allocations for both manual load shed and automatic load shed functions.

System Restoration

System operators maintain blackstart procedures to restore the grid following widespread outages. The integration of large loads challenges traditional assumptions due to their scale and flexibility and the fact that load operators may not have previously interacted with system operators. These procedures must be updated accordingly to ensure that TOPs can reliably blackstart the grid with large loads.

Load Prioritization

In the blackstart process, the system operator defines the initial island parameters, such as the blackstart unit, next start unit, and cranking path. Then the operator plans to connect load to stabilize the island and continue to build out the island adding load, generation, and transmission lines. Prior to system restoration planning and during the restoration process, the system operator is constantly evaluating which loads should be restored first. However, the presence of large loads in the system may impact this prioritization.

As part of the restoration process, TOPs should determine the restoration priority of loads based on their function, size, and location. Large loads may be energy-serving loads (e.g., control centers, loads supporting electric generation, gas production/transmission/distribution with gas compressor stations with high power consumption, critical communication links, or a data center serving applications used by utilities to maintain system reliability) that are integral for the operation of the BPS.¹⁹ TOPs and RCs should identify these loads and how long their backups can sustain them before restoration is needed and include them in the blackstart plans. Priority should be given to these loads to ensure that the BPS can be reliably restarted. This data collection should reflect the best-known knowledge of the load function, size, and location. If these loads are not prioritized in system restoration, then there may be inadequate gas supply to support generation, loss of communication links for system restoration, and the loss of critical infrastructure. To identify these loads, points of contact of large load entities should be provided and updated to the system operators.

In summary, TOPs and RCs should do the following:

- Identify critical large loads and their backup durations
- Reflect best-known data (function, size, location) in restoration plans
- Maintain updated points of contact for large load entities

¹⁹ S. Smillie, M. G. Morgan, and J. Apt, "How vulnerable are US natural gas pipelines to electric outages?," *The Electricity Journal*, vol. 36, no. 2–3, p. 107251, Mar. 2023, doi: <https://doi.org/10.1016/j.tej.2023.107251>.

Load Coordination

During system restoration, TOPs, BAs, and RCs must match load pickup with generation availability and keep load constant and controlled. Uncontrolled load pickup creates large voltage and frequency swings that can negatively impact the stability of blackstart islands and potentially collapse the island. In the past, the TOP would work with the DP to operate distribution equipment (e.g., breakers and switches) to energize load and the load behavior could primarily be safely assumed. With large loads, a single load may be selected for priority restoration and may energize in stages in coordination with the load operator. To accomplish this, the TOP and large load entity should establish communication paths to coordinate during system restoration. Without a communication path, the load cannot be adequately managed for reliable use in blackstart procedures.

While TOs and RCs have traditionally considered the impact of cold load pickup, large loads may exhibit new variations. If large loads are restored early on, they pose a risk to the blackstart island stability as the stability of the island is sensitive to changes in frequency and voltage. To mitigate this, the TO, BA, and RC should establish limits for load variability and blackstart island load ramping in the islands and along the cranking paths and review the priority of large loads in their restoration plans.

Depending on the large load location, characteristics, and controllability, the TOP and RC could potentially use these as a resource for restoration. If TOPs can control the load via dispatch systems or directly communicate with the large load entities (e.g., via large load facility backup power), restoration times could be accelerated compared to picking up conventional distribution load. TOPs and RCs should obtain data on the large load's ability to segment and control its internal site load and maintain a predictable power profile. TOPs and RCs should establish provisions for the use of controllable load resources in blackstart procedures. The large load entity should inform the TOP and RC of potential load ramps during restoration that are driven by the load's internal operations.

Modeling and Simulation for Restoration Studies

To perform accurate studies for selecting blackstart units and cranking paths, it is important for the system operator to obtain information about the load it intends to pick up early in the system restoration process. The TOs, BAs, and RCs should gather the following data on the load and reflect them in the model used for system restoration analysis:

- End-Use Category
- Power Factor
- Controllability
- Ramp Rates
- Backup Power Capability/Power Source/Fuel Reserves
- Ride-Through Capabilities
- Load Variability

Without this information to build the precise model, the system operator is limited in its ability to perform accurate blackstart analyses or create reliable restoration plans.

Given the growing dynamics and uncertainties with these large load behaviors, TOPs should perform detailed restoration studies using transient stability and/or electromagnetic transient simulation tools with appropriate or best-known load models to verify the effectiveness of the restoration plans regularly and adjust their plans to mitigate potential issues.

High-Level Recommendations

The high-level recommendations to maintain system resilience with large load interconnection are summarized in the [Table 8.1](#).

Table 8.1: High-Level Recommendations for Large Load Interconnection regarding System Resilience

#	Recommendation	Applicable Entities	Time Domain
1	Engagement and Communication TOs/DPs should coordinate with PCs to determine if it is appropriate to include the large loads in their UFLS/UVLS programs. TOs/DPs should communicate with large load entities to coordinate on anticipated load shedding before involving large load into the MLS plan. TOPs and load operators should establish communication paths to coordinate during system restoration.	TO/DP, BA, TOP, PC	Planning, Interconnection, Operation
2	Monitoring Real-time monitoring, such as SCADA and telemetry, should be installed for large load that is participating in MLS. UFLS entities should obtain observability to variability of actual UFLS percentage load.	TOP	Interconnection
3	Large Load Splitting TOs/DPs and RCs should explore the partial load shedding guidelines and standards for large load. The TOs/DPs, BA, and RC should develop procedures for partial load restoration and blackstart island ramping criteria for large load.	RC, TO/DP, BA	Interconnection
4	Load Shed Program Review and Coordination Dynamically adjust the frequency of MLS, UVLS, and UFLS to match load growth rate. Applicable entities would be TO, TP, PC. Coordination of UFLS staging between exporting and importing areas can help avoid overstressing transmission between them.	TO/DP, TP, RC	Planning, Operation
5	Study and Analysis TP and PCs should assess frequency stability, including overfrequency performance, the need for overfrequency protection schemes, and the MSSC on the demand side. The TOs/DPs and RCs should perform detailed dynamic restoration studies with appropriate load models to verify the restoration plans regularly and adjust them when necessary.	TO/DP, TP, PC, RC	Planning
6	Large Load Characteristics, Data Collection, and Modeling Data collection of large load should reflect the best-known knowledge of the load function, size, and location at the time of interconnection and be reviewed regularly. TPs should require large load entities to submit the ride-through capability of the load and test report and consider the	TO/DP, TP, PC, RC	Planning, Interconnection

	ride-through characteristics of large load during the contingencies. PCs should consider establishing bounds on allowable variability of loads. TOs and RCs should establish limits for load variability and ramping in the islands and along the cranking paths and review the priority of large loads in their restoration plans. TOs/DPs and RCs should obtain data and confirm the large load's ability to segment and control its internal site load and maintain a predictable power profile. TOs/DPs and RCs should gather critical data on the load and model it appropriately for system restoration analysis.		
8	TOs/DPs and RCs should identify energy serving loads and include them in the blackstart plans.	TO/DP, RC	Operation
9	TOs/DPs and RCs should establish provisions for the use of controllable load resources in blackstart procedures	TO/DP, RC	Interconnection

Chapter 9: Conclusion

In summary, large loads are being integrated faster than ever and at a scale never before seen. New processes will be needed, and existing processes should be updated to mitigate risks from large loads.

Data collection and sharing are foundational to ensuring that accurate assessments can be performed by NERC registered entities. Similar to recent strides in model quality for IBRs, large loads require comprehensive model verification and model validation processes to ensure accurate modeling and reliable operations. This will require strong, ongoing coordination between OEMs, large load entities, and existing NERC registered entities. This also requires dynamic modeling to be accounted for in the earliest stages of design, as opposed to an afterthought once the facility design is complete.

Unless large load entities are registered, TOs alone inherit the burden of creating and enforcing interconnection requirements that account for local and system-wide reliability enhancing processes, including modeling, coordination, analysis, performance, and real-time monitoring. Additionally, new risks and failure modes require TPs and PCs to update their interconnection and planning study processes to ensure that key risks are analyzed and mitigated.

New challenges are being introduced to grid operators, and increased coordination along with new operational requirements can mitigate those challenges. Grid operators will need new data to inform processes that ensure reliable operations, like receiving real-time telemetry and near-term forecasts from large load entities. Additionally, BAs and RCs will need to plan for increased non-conforming variable loads in real-time operations.

Stability risks are clearly demonstrated in prior incident reviews and other analyses. These risks can blackout the power system before operators even have a chance to intervene. Improved data collection, post-event analysis, and dynamic planning studies will be required to prevent critical risks in the stability domain.

Power quality, while currently deemed a lower-likelihood risk, should be accounted for, monitored, and analyzed.

Physical and cyber security, similar to dynamic modeling, must be integrated into facility design at an early stage and coordinated with relevant NERC registered entities.

Finally, load shedding and system restoration risks require consideration at early stages of large load facility design. These risks also require new analysis and coordination from existing NERC registered entities.

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